

ANALYTIC SEWING FOR CHIRAL ALGEBRAS

RAEEZ LORGAT

ABSTRACT. The bar complex of a chiral algebra is a purely algebraic object: it encodes the operator product expansion through residue-contracted point configurations on algebraic curves. Partition functions, by contrast, require convergence of sewing amplitudes across all genera. This paper develops the analytic completion that bridges the two.

We introduce the sewing envelope $\mathcal{A}^{\text{sew}}$, the universal locally convex completion for which every bordism amplitude extends continuously, and formulate a single Hilbert–Schmidt condition on the sector multiplication that implies trace-class closed amplitudes at all genera. For the Heisenberg algebra we prove that the sewing envelope is $\text{Sym}A^2(D)$, the pair-of-pants amplitude is the second quantisation of a one-particle Bergman restriction, and every genus- g partition function is a Fredholm determinant; this is the decisive first case. For lattice vertex algebras we prove that the partition function factors into a Heisenberg Fredholm determinant times a Siegel theta function, both absolutely convergent on the Siegel upper half-space. The general HS-sewing theorem verifies the criterion for the entire standard landscape: Heisenberg, affine Kac–Moody, Virasoro, $\mathcal{W}_N$ algebras, and lattice vertex algebras.

We identify a three-layer analytic gap separating the algebraic bar construction from convergent partition functions: the sewing envelope for interacting algebras, metric independence of the ind-Hilbert factorisation, and the coderived shadow at genus $g \geq 1$. The coderived category D^{co} is the correct receptacle: curved differentials ($d^2 = \kappa \cdot \omega_g \neq 0$) are permitted, coacyclic complexes replace acyclic ones, and the BV/bar comparison holds for all four shadow classes including class M (Virasoro, $\mathcal{W}_N$), where the naive chain-level identification fails.

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Date: April 23, 2026.

2020 Mathematics Subject Classification. 17B69 (primary); 14H10, 30H20, 47B10, 81T40 (secondary).

Key words and phrases. Chiral algebras, sewing, bar construction, Hilbert–Schmidt operators, Bergman space, Fredholm determinant, coderived category, partition functions.

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1. INTRODUCTION

1.1. The algebraic bar complex and the convergence problem. The bar complex of a chiral algebra $\mathcal{A}$ is the tensor coalgebra $B(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with differential extracted from the operator product expansion by residues at collision diagonals in the Fulton–MacPherson compactification $\overline{\text{FM}}_n(\mathbb{C})$. The construction is purely algebraic: it uses only the OPE coefficients, the augmentation ideal $\bar{\mathcal{A}} = \ker(\varepsilon)$, and the desuspension s^{-1} with $|s^{-1}v| = |v| - 1$. The bar differential satisfies $D^2 = 0$ on a fixed curve, and the Maurer–Cartan element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0 \in \text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}})$ governs the modular obstruction tower.

Partition functions require more. A genus- g partition function is a trace

$$Z_g(\mathcal{A}; \Omega) = \text{Tr}_{\mathcal{A}}(q_1^{L_0 - c/24} \cdots q_g^{L_0 - c/24}) \quad (1.1)$$

over a state space completed in a topology for which the operator q^{L_0} is trace class. The algebraic bar complex produces formal power series in the sewing parameters; the question is whether these series converge to honest complex numbers.

This convergence problem is the analytic gap at the heart of the modular Koszul duality programme [5]. The five theorems (A through D and H) of that programme are stated and proved algebraically; the analytic sewing package developed here is the mechanism by which the algebraic skeleton extends to convergent invariants.

1.2. Main results. We prove five results.

The sewing envelope. For any positive-energy chiral algebra $\mathcal{A}$, we construct the *sewing envelope* $\mathcal{A}^{\text{sew}}$ (Definition 2.1), the universal locally convex completion for which every bordism amplitude extends continuously, and establish its universal property (Proposition 2.2).

The Hilbert–Schmidt sewing condition. We formulate a single analytic hypothesis (Definition 3.1): the sector multiplication operators $m_{a,b}^c: H_a \otimes H_b \rightarrow H_c$ satisfy

$$\sum_{a,b,c \geq 0} q^{a+b+c} \|m_{a,b}^c\|_{\text{HS}}^2 < \infty \quad (1.2)$$

for some $0 < q < 1$. We prove (Proposition 3.2) that this single estimate delivers the full sewing package: every closed amplitude is trace class on the weighted completion $H_q := \bigoplus_n q^n H_n$.

The Heisenberg Fredholm determinant. For the Heisenberg algebra $\mathcal{H}_k$ (Theorem 4.3): the sewing envelope is $\text{Sym} A^2(D)$; the pair-of-pants amplitude is the second quantisation $\Gamma(R)$ of a one-particle Bergman restriction; every genus- g partition function is the Fredholm determinant

$$Z_g(\mathcal{H}_k; \Omega) = (\det \text{Im } \Omega)^{-k/2} \cdot (\det'_\zeta \Delta_{\Sigma_g})^{-k/2}. \quad (1.3)$$

Lattice sewing. For a lattice vertex algebra V_Λ of rank r (Theorem 5.1): the genus- g partition function factors as

$$Z_g(V_\Lambda; \Omega) = Z_g(\mathcal{H}_r; \Omega) \cdot \Theta_\Lambda(\Omega), \quad (1.4)$$

where $\Theta_\Lambda(\Omega) = \sum_{\lambda \in \Lambda^g} e^{i\pi \lambda^T \Omega \lambda}$ is the Siegel theta function of the lattice, absolutely convergent on the Siegel upper half-space $\mathfrak{H}_g$.

The general HS-sewing theorem. Any positive-energy chiral algebra with subexponential sector growth and polynomial OPE coefficient growth satisfies the HS-sewing condition (Theorem 6.1). The criterion verifies for the Heisenberg, affine Kac–Moody at non-critical level, Virasoro at every central charge, $\mathcal{W}_N$ algebras, and lattice vertex algebras.

1.3. The three-layer gap and the coderived category. Convergent partition functions are the beginning of the analytic theory, not its end. Three layers of structure separate the algebraic bar construction from a complete analytic modular Koszul programme:

- (1) *The sewing envelope for interacting algebras.* For the Heisenberg and lattice algebras, the sewing envelope admits an explicit description (Bergman symmetric algebra, charge-refined Hilbert completion). For non-abelian chiral algebras (affine Kac–Moody at generic level, Virasoro, $\mathcal{W}_N$), the sewing envelope exists abstractly but lacks a concrete Hilbert-space identification.
- (2) *Metric independence of the ind-Hilbert factorisation.* The ind-Hilbert completion of Moriwaki [6] depends a priori on a conformal metric on the curve; the algebraic bar complex does not. Identifying the two requires proving that the analytic completion is independent of the metric, a functional-analytic problem with no algebraic counterpart.
- (3) *The coderived shadow at $g \geq 1$.* At genus $g \geq 1$ the bar complex carries curvature: $d_{\text{hb}}^2 = \kappa(\mathcal{A}) \cdot \omega_g \neq 0$. The ordinary derived category is empty (every object is acyclic). The correct receptacle

is the coderived category D^{co} , defined by totalising the curvature bicomplex.

In Sections 7 and 8 we develop the third layer in detail: the coderived category as the natural home for curved bar complexes, the notion of coacyclicity replacing acyclicity, and the BV/bar comparison theorem in D^{co} for all four shadow classes (G, L, C, M) of the standard landscape.

1.4. Relation to prior work. Segal [9] formulated the sewing axiom as the defining property of a conformal field theory. Zhu [10] proved modular invariance of genus-1 characters for rational vertex operator algebras. Huang [4] extended the sewing of sphere amplitudes to genus one and by induction to all genera, under a list of analytic convergence hypotheses. The HS criterion of this paper is strictly stronger: a single norm estimate (1.2) implies all convergences simultaneously.

Moriwaki [6] identified the Heisenberg sewing envelope with $\text{Sym}A^2(D)$ through factorisation homology valued in ind-Hilbert spaces. We reinterpret this as a one-particle reduction of the pair-of-pants operator and deduce the Fredholm determinant formula (1.3).

The Belavin–Knizhnik theorem [2] expresses the bosonic string measure as a product of determinants of Laplacians; our formula (1.3) is the vertex-algebraic incarnation. Beilinson–Drinfeld [1] develop the algebraic theory of chiral algebras and factorisation; our bar complex is the explicit chain model of their factorisation coalgebra.

The coderived category for curved dg-modules was introduced by Positselski [7]; its application to curved bar complexes in the chiral setting appears in [5]. The BV/bar identification in the coderived category (Theorem 8.7) extends the genus-0 chain-level result of Costello and Gwilliam [3] to all genera.

1.5. The modular Koszul duality context. The analytic sewing programme is one component of a larger algebraic structure. The modular Koszul duality programme of [5] constructs, for every chirally Koszul algebra $\mathcal{A}$, a package of algebraic invariants:

- (a) the bar complex $B(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with differential and deconcatenation coproduct;
- (b) the Maurer–Cartan element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0 \in \text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}})$;
- (c) the shadow obstruction tower $\Theta_{\mathcal{A}} = \sum_{r \geq 2} S_r(\mathcal{A})$ with $S_2 = \kappa(\mathcal{A})$ the modular characteristic;
- (d) the Koszul dual $\mathcal{A}^!$ and the complementarity relation $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = K(\mathcal{A})$;
- (e) five theorems (A through D and H) controlling the structure of the bar complex, bar-cobar inversion, complementarity, the genus expansion, and chiral Hochschild cohomology.

All of this is purely algebraic: the bar complex is built from OPE coefficients and residues; no convergence hypothesis enters. The analytic sewing

programme begins where the algebraic programme ends: it asks when the formal genus expansion $\Theta_{\mathcal{A}} = \sum_g \Theta_{\mathcal{A}}^{(g)}$ converges to a well-defined partition function $Z_g(\mathcal{A}; \Omega)$.

The algebraic programme establishes the formula; the analytic programme establishes the number. The two are connected by the HS-sewing condition, which is precisely the analytic hypothesis that converts the algebraic genus expansion into a convergent Fredholm expansion.

Remark 1.1 (The four shadow classes and sewing). The shadow classification (G, L, C, M) of the standard landscape controls the complexity of the sewing programme:

- (a) *Class G* (Heisenberg, lattice): the shadow tower terminates at degree 2; the sewing programme is fully explicit (Sections 4 and 5).
- (b) *Class L* (affine Kac–Moody): the shadow tower terminates at degree 3; the sewing programme is proved abstractly (HS-sewing, Theorem 6.1) but the sewing envelope is not yet identified concretely.
- (c) *Class C* ($\beta\gamma$, bc): the shadow tower terminates at degree 4; the BV/bar comparison is conditional on harmonic decoupling.
- (d) *Class M* (Virasoro, $\mathcal{W}_N$): the shadow tower is infinite; the chain-level BV/bar comparison fails; the coderived category is essential (Section 8).

1.6. Outline. Section 2 constructs the sewing envelope. Section 3 formulates the Hilbert–Schmidt sewing condition and proves the trace-class theorem. Section 4 treats the decisive first case: the Heisenberg Fredholm determinant. Section 5 develops lattice sewing via theta functions. Section 6 proves the general HS-sewing theorem and verifies the criterion on the standard landscape. Section 7 analyses the three-layer analytic gap. Section 8 develops the coderived category as the correct receptacle for curved bar complexes at $g \geq 1$.

1.7. Conventions. All chiral algebras are $\mathbb{Z}_{\geq 0}$ -graded (positive energy), with finite-dimensional weight spaces. Grading is cohomological: $|d| = +1$. Desuspension lowers degree: $|s^{-1}v| = |v| - 1$. The bar complex uses the augmentation ideal: $B(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with $\bar{\mathcal{A}} = \ker(\varepsilon)$. The Dedekind eta function is $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$ with $q = e^{2\pi i \tau}$.

2. THE SEWING ENVELOPE

2.1. Algebraic cores and bordism amplitudes. An *algebraic chiral core* is a $\mathbb{Z}_{\geq 0}$ -graded vector space $\mathcal{A}_{\text{alg}} = \bigoplus_{n \geq 0} H_n$ equipped with the operator product expansion of a vertex operator algebra of positive energy, with $H_0 = \mathbb{C} \cdot \mathbf{1}$ and $\dim H_n < \infty$. The sector components H_n are the conformal-weight eigenspaces. The augmentation $\varepsilon: \mathcal{A}_{\text{alg}} \rightarrow \mathbb{C}$ is the projection to H_0 . The augmentation ideal $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the input to the bar construction $B(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$.

A finite conformally flat bordism is a compact Riemann surface Σ with parametrised in/out boundary collars. Each such Σ determines an algebraic amplitude

$$A_{\Sigma}^{\text{alg}}: \mathcal{A}_{\text{alg}}^{\otimes m} \longrightarrow \mathcal{A}_{\text{alg}}^{\otimes \ell},$$

a priori valued in formal series of conformal-weight components. Matrix coefficients

$$\langle \eta, A_{\Sigma}^{\text{alg}}(\xi_1 \otimes \cdots \otimes \xi_m) \rangle \in \mathbb{C}[[\text{moduli of } \Sigma]]$$

are the objects whose convergence the sewing programme must establish.

2.2. Construction of the sewing envelope.

Definition 2.1 (Sewing envelope). For every finite conformally flat bordism Σ and every choice of boundary vectors $\xi_i \in \mathcal{A}_{\text{alg}}$, $\eta \in \mathcal{A}_{\text{alg}}^{\vee}$, define the seminorm

$$p_{\Sigma, \xi, \eta}(a) = |\langle \eta, A_{\Sigma}^{\text{alg}}(\xi_1 \otimes \cdots \otimes a \otimes \cdots \otimes \xi_m) \rangle|. \quad (2.1)$$

The *sewing envelope* $\mathcal{A}^{\text{sew}}$ is the Hausdorff completion of $\mathcal{A}_{\text{alg}}$ for the locally convex topology generated by all seminorms $p_{\Sigma, \xi, \eta}$.

The sewing envelope is the smallest completion that remembers every local-to-global operation at once: it is tautologically the correct target for any analytic bootstrap.

Proposition 2.2 (Universal property). *Let E be any complete locally convex realisation of $\mathcal{A}_{\text{alg}}$ for which every algebraic amplitude A_{Σ}^{alg} extends continuously. Then the identity on $\mathcal{A}_{\text{alg}}$ extends uniquely to a continuous map $\mathcal{A}^{\text{sew}} \rightarrow E$.*

Proof. By hypothesis, each seminorm $p_{\Sigma, \xi, \eta}$ is continuous on E . Hence the identity $\mathcal{A}_{\text{alg}} \rightarrow E$ is continuous for the sewing topology. By completeness of E , it extends uniquely from the Hausdorff completion $\mathcal{A}^{\text{sew}}$. $\square$

Remark 2.3 (Comparison with other completions). The sewing envelope need not coincide with naive completions: the q -adic completion $\widehat{\bigoplus_n q^n H_n}$, the Hilbert-space completion with respect to a fixed invariant form, Zhu's C_2 -quotient, or Moriwaki's ind-Hilbert completion. All of these are quotients or subspaces of $\mathcal{A}^{\text{sew}}$ under the universal property, but none is a priori large enough to carry every bordism amplitude as a continuous map. The HS-sewing criterion of the next section is the device by which one identifies a concrete Hilbert completion $H_q \subset \mathcal{A}^{\text{sew}}$.

2.3. Functoriality. The sewing envelope construction is functorial in the algebraic core. A morphism of vertex algebras $f: \mathcal{A} \rightarrow \mathcal{B}$ that preserves the conformal grading and the OPE induces a continuous map $f^{\text{sew}}: \mathcal{A}^{\text{sew}} \rightarrow \mathcal{B}^{\text{sew}}$. In particular, any subalgebra inclusion $\mathcal{A} \hookrightarrow \mathcal{B}$ (such as the inclusion of a Heisenberg subalgebra into a lattice vertex algebra) induces a continuous inclusion of sewing envelopes.

Proposition 2.4 (Direct-sum decomposition). *Let $\mathcal{A} = \bigoplus_{\gamma \in I} \mathcal{A}_\gamma$ be a decomposition of the algebraic core into sectors preserved by the pair-of-pants multiplication. Then*

$$\mathcal{A}^{\text{sew}} = \widehat{\bigoplus_{\gamma \in I} \mathcal{A}_\gamma^{\text{sew}}},$$

where the completion is in the product topology.

Proof. Each seminorm $p_{\Sigma, \xi, \eta}$ decomposes by sector selection (insertion of projection operators onto the sector γ at each boundary circle); the cross-terms are separate seminorms. The completion of a direct sum in a product of locally convex topologies is the product of completions. $\square$

2.4. The five levels of analytic structure. The sewing envelope sits at the fourth of five levels of analytic structure for chiral algebras, arranged in a strict chain of implications:

Theorem 2.5 (Five-level hierarchy). *The following conditions on a positive-energy chiral algebra $\mathcal{A}$ form a strict chain:*

- (i) **(Algebraic)** $D_{\mathcal{A}}^2 = 0$ on the bar complex. *Universal: holds for every chiral algebra.*
- (ii) **(Coderived)** *The homotopy transfer theorem applies weight-by-weight in the coderived category D^{co} . Requires: positive energy ($\dim H_n < \infty$).*
- (iii) **(Koszul)** *The curved bar complex and the flat model are linked by the coderived-to-derived comparison. Requires: PBW collapse at the E_2 page of the bar spectral sequence.*
- (iv) **(Analytic)** *The genus- g partition functions converge absolutely. Requires: HS-sewing (polynomial OPE growth, subexponential sector growth).*
- (v) **(Modular functor)** *Conformal blocks are finite-dimensional and the modular group acts. Requires: C_2 -cofiniteness.*

The first four levels form the chain (iv) $\subsetneq$ (iii) $\subsetneq$ (ii) $\subsetneq$ (i). Level (v) implies (iv), (ii), and (i), but its relation to (iii) is not one of containment: C_2 -cofinite simple quotients $L_k(\mathfrak{g})$ at admissible level satisfy (v) but may fail (iii), while Koszul universal algebras $V^k(\mathfrak{g})$ at generic level satisfy (iii) but not (v).

Proof sketch. Level (i): $D^2 = 0$ holds because the bar differential is the Feynman-transform differential of a modular operad algebra, and $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$ implies $D^2 = 0$. Level (ii): the positive-energy axiom gives weight-by-weight finite-dimensional SDR data for the homotopy transfer. Level (iii): the PBW collapse gives diagonal Ext vanishing. Level (iv): HS-sewing (Theorem 6.1). Level (v): Zhu's theorem [10] gives finite-dimensional conformal blocks and modular group action.

The inclusions are strict. For (i) $\neq$ (ii): a chiral algebra without positive energy satisfies $D^2 = 0$ but lacks the SDR. For (ii) $\neq$ (iii): admissible-level quotients where bar-Ext differs from ordinary-Ext. For (iii) $\neq$ (iv): a chirally

Koszul algebra with super-polynomial OPE growth. For (iv) $\neq$ (v): the universal $\mathcal{W}$ -algebra $V^k(\mathfrak{g})$ at generic irrational level. $\square$

Remark 2.6 (The standard landscape). The standard landscape (Heisenberg, affine Kac–Moody at non-critical level, Virasoro, $\mathcal{W}_N$, $\beta\gamma$ and bc systems, lattice vertex algebras) satisfies levels (i) through (iv). Those families that are also C_2 -cofinite (simple quotients at admissible levels, self-dual lattice VOAs) satisfy all five levels. The sewing envelope is the universal object at level (iv); the present paper develops this level in detail.

2.5. The bar complex as a family over moduli. The bar complex at genus g is a family of cochain complexes parametrised by $\overline{\mathcal{M}}_g$. On the smooth locus $\mathcal{M}_g$ the differential satisfies $d^2 = 0$; on the compactification $\overline{\mathcal{M}}_g$ the differential acquires curvature $d_{\text{fib}}^2 = \kappa \cdot \omega_g$ from the boundary strata.

The Gauss–Manin connection $\nabla^{\text{GM}}: B^{(g)}(\mathcal{A}) \rightarrow B^{(g)}(\mathcal{A}) \otimes \Omega_{\overline{\mathcal{M}}_g}^1$ controls the variation of the bar complex with moduli. The total differential $D_g = d_{\text{fib}} + \nabla^{\text{GM}}$ satisfies $D_g^2 = 0$: the fiberwise curvature $d_{\text{fib}}^2 = \kappa\omega_g$ is exactly cancelled by the Gauss–Manin curvature $(\nabla^{\text{GM}})^2 = -\kappa\omega_g$.

The analytic sewing programme extends this family to a family of Hilbert spaces: on each fiber (a fixed Riemann surface), the algebraic bar complex is completed to a Hilbert space by the HS-sewing condition, and the Gauss–Manin connection extends continuously to the completed fibers.

Proposition 2.7 (Continuity of the Gauss–Manin connection). *Under HS-sewing, the Gauss–Manin connection ∇_g^{GM} extends to a bounded operator on the weighted completion H_q for each g , and the total differential $D_g = d_{\text{fib}} + \nabla_g^{\text{GM}}$ satisfies $D_g^2 = 0$ on the completed bar complex.*

Proof. The Gauss–Manin connection acts on bar elements by variation of the bar propagator $d \log E(z, w)$ with respect to moduli. The variation is a holomorphic $(1, 0)$ -form on $\overline{\mathcal{M}}_g$ valued in operators on H_q ; the HS bound on the pair-of-pants multiplication controls the operator norm of these variations, because each variation is expressed as a residue of the bar differential applied to a collar region. The identity $D_g^2 = 0$ is algebraic (it holds on the dense algebraic core) and extends by continuity. $\square$

3. THE HILBERT–SCHMIDT SEWING CONDITION

The universal sewing envelope of Definition 2.1 is an abstract object. To compute with it one needs a concrete Hilbert completion on which the sewing operators are not merely defined but small enough in norm to compose.

3.1. The condition.

Definition 3.1 (HS-sewing). Let $\mathcal{A}_{\text{alg}} = \bigoplus_n H_n$ be an algebraic chiral core with a pre-Hilbert structure on each sector (for example, the restriction of the invariant bilinear form to H_n). Write the pair-of-pants sector multiplications as

$$m_{a,b}^c: H_a \otimes H_b \longrightarrow H_c.$$

Say $\mathcal{A}$ satisfies *HS-sewing* at parameter $q \in (0, 1)$ if

$$\sum_{a,b,c \geq 0} q^{a+b+c} \|m_{a,b}^c\|_{\text{HS}}^2 < \infty. \quad (3.1)$$

The *weighted completion* is $H_q := \widehat{\bigoplus_{n \geq 0} q^n H_n}$, with inner product $\langle v, w \rangle_q = q^{2n} \langle v, w \rangle_{H_n}$ for $v, w \in H_n$.

The scaling q^{2n} is the unique convention under which the weighted pair-of-pants m_q is Hilbert–Schmidt precisely when (3.1) holds.

3.2. The trace-class theorem.

Proposition 3.2 (Closed amplitudes are trace class). *Suppose $\mathcal{A}$ satisfies HS-sewing at parameter q . Then:*

- (i) *the weighted pair-of-pants $m_q: H_q \otimes H_q \rightarrow H_q$ is Hilbert–Schmidt, with*

$$\|m_q\|_{\text{HS}}^2 = \sum_{a,b,c} q^{a+b+c} \|m_{a,b}^c\|_{\text{HS}}^2;$$

- (ii) *every closed amplitude of $\mathcal{A}$, built from any finite composition of pair-of-pants multiplications and traces, is trace class on H_q ;*
- (iii) *the sewing envelope $\mathcal{A}^{\text{sew}}$ admits a continuous linear map $\mathcal{A}^{\text{sew}} \rightarrow H_q$ with dense image.*

Proof. Claim (i): identifying $H_a \otimes H_b$ with the weight- $(a+b)$ subspace of $H_q^{\otimes 2}$, the sector component $m_{a,b}^c$ contributes $q^a \cdot q^b \cdot q^{-c} \cdot q^{2c} = q^{a+b+c}$ to the Hilbert–Schmidt norm of m_q (the factor q^{-c} comes from the output side, the factor q^{2c} from the normalisation of the inner product on H_c). Summing gives $\|m_q\|_{\text{HS}}^2 = \sum q^{a+b+c} \|m_{a,b}^c\|_{\text{HS}}^2 < \infty$.

Claim (ii): a genus- g Riemann surface with n punctures admits a pants decomposition into $2g-2+n$ pairs of pants. Each internal sewing circle corresponds to a Hilbert–Schmidt pair-of-pants composition, and the composition of two Hilbert–Schmidt operators is trace class. Every closed amplitude (no free external legs) evaluates to the trace of a finite composition of trace-class operators.

Claim (iii): the seminorms $p_{\Sigma, \xi, \eta}$ of Definition 2.1 are all dominated by the H_q inner product, because each matrix coefficient factorises through the trace-class operator of claim (ii). Continuity of the identity $\mathcal{A}_{\text{alg}} \rightarrow H_q$ for the sewing topology follows. $\square$

Corollary 3.3 (Partition functions converge). *Under HS-sewing at parameter q , the genus- g partition function $Z_g(\mathcal{A}; \Omega) = \text{Tr}(q_1^{L_0-c/24} \dots q_g^{L_0-c/24})$ converges absolutely for all period matrices $\Omega \in \mathfrak{H}_g$ with $|q_j| = |e^{2\pi i \Omega_{jj}}| < q$, producing a holomorphic function on a domain in the Siegel upper half-space.*

Proof. The partition function is a closed amplitude (trace over the state space with g sewing circles). By Proposition 3.2(ii), it is the trace of a trace-class operator on H_q , hence absolutely convergent. Holomorphicity

in the moduli follows from the absolute convergence and the holomorphic dependence of the sewing kernel on Ω . $\square$

Corollary 3.4 (Gram positivity). *Under HS-sewing, the connected sewing amplitudes $a_{\mathcal{A}}(N) \in \mathbb{R}_{\geq 0}$ define a positive-semidefinite inner product on generating Dirichlet polynomials.*

Proof. Trace-class operators form a two-sided ideal with a well-defined trace, positive when applied to a positive operator; the sewing amplitude is a composition of positive pair-of-pants operators, and positivity passes through. $\square$

3.3. The sewing Gram matrix and its determinant. The Hilbert–Schmidt bound controls not only convergence but spectral asymptotics. Let K_g denote the sewing operator (composition of pair-of-pants restrictions around a cycle of the genus- g surface). The connected free energy is

$$F_g^{\text{conn}}(\mathcal{A}; q) = -\log \det(1 - K_g), \quad (3.2)$$

convergent whenever $\|K_g\|_1 < 1$. The eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ of K_g satisfy $\sum_j \lambda_j < \infty$ (trace class), and the Fredholm determinant $\det(1 - K_g) = \prod_j (1 - \lambda_j)$ converges absolutely. For the Heisenberg, the eigenvalues are q^n ($n \geq 1$) and the Fredholm determinant is the Dedekind eta function (see Section 4).

3.4. The HS-sewing condition and the shadow classes. The HS-sewing condition interacts with the shadow classification (G, L, C, M) of the modular Koszul duality programme. The shadow class determines the OPE pole structure and hence the polynomial degree N in the OPE growth bound.

Proposition 3.5 (Shadow class and OPE growth). *The OPE growth exponent N in the bound $|C_{a,i;b,j}^{c,k}| \leq K(a+b+c+1)^N$ is controlled by the shadow class:*

Class	Families	N	Mechanism
G	Heisenberg, free fields	1	quadratic OPE
L	affine Kac–Moody	1	linear current algebra
C	$\beta\gamma, bc$	1	first-order systems
M	Virasoro, $\mathcal{W}_N$	$2N$	higher-spin fields

Classes G, L, and C all have $N = 1$ (linear OPE growth); class M has N growing with the number of generators.

Proof. For class G: the OPE $J(z)J(w) \sim k/(z-w)^2$ has no field on the right-hand side beyond the central term; the structure constants are bounded by the level k , independent of weight.

For class L: the OPE $J^a(z)J^b(w) \sim k\delta^{ab}/(z-w)^2 + f^{ab}_c J^c(w)/(z-w)$ has at most one derivative insertion, giving $|C| \leq K(n+1)$.

For class C: the $\beta\gamma$ system has OPE $\beta(z)\gamma(w) \sim 1/(z-w)$, a simple pole; all higher correlators factor through this pairing by Wick's theorem.

For class M: the Virasoro OPE is quartic, and normal-ordered products of T with itself produce structure constants growing as $(n+1)^2$. For $\mathcal{W}_N$, the spin- s generator contributes $2s$ poles, and the OPE growth is bounded by $(n+1)^{2N}$. $\square$

The shadow class also determines the convergence rate of the HS-sewing sum. For classes G, L, C (all with $N = 1$), the HS-sewing sum converges rapidly:

$$\sum_{a,b,c} q^{a+b+c} \|m_{a,b}^c\|_{\text{HS}}^2 \leq 9K^2 \left(\sum_n q^n \dim H_n (n+1)^2 \right) \left(\sum_n q^n \dim H_n \right)^2,$$

and the rate-limiting factor is the sector growth $\dim H_n$. For class M with large N , the polynomial factor $(n+1)^{2N}$ slows convergence but does not affect the qualitative conclusion (the exponential decay of q^n dominates any polynomial growth).

3.5. Quantitative HS bounds. The HS-sewing condition provides not just qualitative convergence but quantitative bounds on partition functions.

Proposition 3.6 (Partition function bound). *Let $\mathcal{A}$ satisfy HS-sewing at parameter q . Then the genus- g partition function satisfies*

$$|Z_g(\mathcal{A}; \Omega)| \leq (\|m_q\|_{\text{HS}})^{2(2g-2)} \quad (3.3)$$

for all $\Omega \in \mathfrak{H}_g$ with $|q_j| < q$.

Proof. A pants decomposition of the genus- g surface has $2g-2$ pairs of pants and $3g-3$ sewing circles. Each pair of pants contributes a factor of $\|m_q\|_{\text{HS}}$ (the Hilbert–Schmidt norm of the pair-of-pants multiplication); each sewing circle composes two such factors. The closed amplitude (the trace) is bounded by the product of all HS norms: $|Z_g| \leq \|m_q\|_{\text{HS}}^{2(2g-2)}$ (the exponent counts the number of internal pair-of-pants composition operations in a genus- g surface with no punctures). $\square$

Corollary 3.7 (Growth in genus). *Under HS-sewing, the partition functions grow at most exponentially in genus: $\log |Z_g| \leq (2g-2) \log \|m_q\|_{\text{HS}}^2$. For the Heisenberg, $\|m_q\|_{\text{HS}}^2 = k \sum_n q^{2n+1}/(1-q)^2$, and the growth rate is controlled by the level k and the sewing parameter q .*

This exponential-in-genus bound is consistent with the predictions of the Belavin–Knizhnik factorisation [2] and with the genus- g Faber–Pandharipande evaluation λ_g^{FP} , which grows as $(2g)!/(2\pi)^{2g}$.

3.6. Optimality of the HS condition. The HS-sewing condition is not sharp: it is a sufficient condition for convergent partition functions, not a necessary one. Two weaker conditions that also imply convergence in special cases:

- (a) *C₂-cofiniteness*: Zhu's condition [10] implies finite-dimensional conformal blocks and hence convergent characters. This is strictly stronger than HS-sewing for rational VOAs (it implies modularity, which HS-sewing does not), but strictly weaker in scope (it applies only to C₂-cofinite algebras, while HS-sewing applies to generic-level universal algebras).
- (b) *Trace-class pair-of-pants*: one could replace the Hilbert–Schmidt condition on $m_{a,b}^c$ with a trace-class condition on the composed operator $m_q \circ m_q^*$. This is strictly weaker than HS-sewing (trace class $\subset$ Hilbert–Schmidt) but harder to verify from OPE data alone.

The HS condition strikes the optimal balance between generality (it applies to the full standard landscape) and verifiability (it can be checked from character formulas and OPE tables).

4. THE HEISENBERG SEWING: FREDHOLM DETERMINANT

The Heisenberg vertex algebra $\mathcal{H}_k$ is the decisive first case: a chiral algebra for which the entire analytic sewing programme admits a complete, explicit description. Every step reduces to Fock-space combinatorics and Bergman-space functional analysis.

4.1. The Heisenberg algebra. The Heisenberg vertex algebra $\mathcal{H}_k$ at level k is generated by a single current $J(z)$ with operator product expansion

$$J(z)J(w) \sim \frac{k}{(z-w)^2}.$$

The sectors H_n are spanned by monomials $a_{-n_1-1} \cdots a_{-n_j-1} \mathbf{1}$ with $\sum (n_i + 1) = n$; the sector generating function is $\sum_{n \geq 0} \dim H_n q^n = \prod_{n \geq 1} (1 - q^n)^{-1}$, equivalently $q^{1/24}/\eta(q)$. The Heisenberg is the simplest member of the standard landscape, class G in the shadow classification (the shadow tower terminates at degree 2), with $\kappa(\mathcal{H}_k) = k$.

4.2. The Heisenberg OPE and HS-sewing verification. The HS-sewing condition for the Heisenberg is verified directly from the OPE. The pair-of-pants multiplication $m_{a,b}^c: H_a \otimes H_b \rightarrow H_c$ is nonzero only when $c \leq a + b$ (the weight cannot increase beyond the total input weight). The OPE $J(z)J(w) \sim k/(z-w)^2$ produces structure constants bounded by k : the n -th mode a_n satisfies $[a_m, a_n] = km\delta_{m+n,0}$, so the only nontrivial sector multiplication is the Wick contraction $m_{a,b}^{a+b-1}: a_{-a-1} \mathbf{1} \otimes a_{-b-1} \mathbf{1} \mapsto k(a+1)\delta_{b,0} a_{-a-1} \mathbf{1}$ (plus normal-ordered products). The Hilbert–Schmidt norm is

$$\|m_{a,b}^c\|_{\text{HS}}^2 \leq k^2 p(a)p(b)p(c)(a+b+c+1)^2,$$

and the HS-sewing sum is

$$\sum_{a,b,c} q^{a+b+c} \|m_{a,b}^c\|_{\text{HS}}^2 \leq 9k^2 \left(\sum_n q^n p(n)(n+1)^2 \right) \left(\sum_n q^n p(n) \right)^2.$$

Each series converges for $0 < q < 1$ because $p(n)$ grows subexponentially ($\log p(n) \sim \pi\sqrt{2n/3}$) while q^n decays exponentially. The radius of convergence $q = 1$ is sharp: at $q = 1$ the series $\sum p(n) = \infty$.

Remark 4.1 (The partition number and subexponential growth). The Hardy–Ramanujan asymptotic $p(n) \sim \frac{1}{4n\sqrt{3}} \exp(\pi\sqrt{2n/3})$ shows that $\log p(n) = O(\sqrt{n})$, hence $p(n) \leq C \exp(\alpha\sqrt{n})$ for appropriate C and α . The exponential decay $q^n = \exp(-n \log(1/q))$ dominates the subexponential growth for any $q < 1$: $q^n p(n) \leq C \exp(\alpha\sqrt{n} - n \log(1/q))$, which decays superpolynomially. This is the mechanism behind the HS-sewing convergence for all standard families.

4.3. The Bergman space. Let $A^2(D)$ be the Bergman space of square-integrable holomorphic functions on the unit disk $D \subset \mathbb{C}$, with inner product $\langle f, g \rangle = \int_D f \bar{g} dA$ and orthonormal basis $e_n(z) = \sqrt{(n+1)/\pi} z^n$ for $n \geq 0$. The symmetric algebra $\text{Sym} A^2(D)$ is the bosonic Fock space built on the one-particle Bergman space. The Bergman reproducing kernel is

$$K_D(z, w) = \frac{1}{\pi(1 - z\bar{w})^2}. \quad (4.1)$$

Definition 4.2 (Mode–Bergman map). The *mode–Bergman map* $\Theta: \mathcal{H}_k^{\text{alg}} \rightarrow \text{Sym} A^2(D)$ is the dense linear map defined on the algebraic Fock core by

$$\begin{aligned} \Theta(a_{-n-1} \mathbf{1}) &= \sqrt{n+1} z^n, \\ \Theta(a_{-n_1-1} \cdots a_{-n_j-1} \mathbf{1}) &= \text{Sym}(\Theta(a_{-n_1-1} \mathbf{1}) \otimes \cdots \otimes \Theta(a_{-n_j-1} \mathbf{1})). \end{aligned}$$

The map Θ intertwines the algebraic Fock space grading with the polynomial-degree grading of the Bergman space and sends the invariant bilinear form on $\mathcal{H}_k$ to the Bergman inner product (up to the level factor k).

4.4. The Heisenberg sewing theorem.

Theorem 4.3 (Heisenberg sewing). *Let $\mathcal{H}_k$ be the Heisenberg vertex algebra at level k and let Θ be the mode–Bergman map.*

- (i) *The sewing envelope $\mathcal{H}_k^{\text{sew}}$ is identified via Θ with $\text{Sym} A^2(D)$.*
- (ii) *The completed bar differential on Bergman tensors is the closure of the Gaussian collision operator.*
- (iii) *The pair-of-pants sewing amplitude is the second quantisation $\Gamma(R)$ of a one-particle Bergman restriction $R: A^2(D_{\text{out}}) \rightarrow A^2(D_{\text{in},1}) \otimes A^2(D_{\text{in},2})$.*
- (iv) *Every closed genus- g amplitude is a Fredholm determinant on the reduced one-particle Bergman space $A_0^2(\partial D)$ (constant mode removed).*

Proof. Claim (i) is the ind-Hilbert identification of Moriwaki [6], who constructs a conformally flat 2-disk algebra structure on $\text{Sym} A^2(D)$ isomorphic to the Heisenberg sewing completion by factorisation homology valued in IndHilb . The universal property (Proposition 2.2) identifies his completion with $\mathcal{H}_k^{\text{sew}}$.

Claim (ii) follows from the analytic-algebraic comparison: the algebraic bar differential, in the monomial basis of the Fock space, is

$$d_{\text{bar}}(a_{-n-1}\mathbf{1} \otimes a_{-m-1}\mathbf{1}) = k \delta_{n+m+2,0} \mathbf{1}$$

(the simple-pole Wick contraction of the generating current). Under Θ , this contraction is the Gaussian collision operator $G(z, w) = k/(1 - z\bar{w})^2$ on $A^2(D) \otimes A^2(D)$, the reproducing kernel of the Bergman space.

Claims (iii) and (iv) follow from the one-particle sewing theorem below. $\square$

4.5. One-particle sewing and the Fredholm determinant. The Heisenberg amplitude factorises completely through a one-particle operator. This is the analytic avatar of Wick's theorem.

Theorem 4.4 (One-particle sewing). *The Heisenberg pair-of-pants amplitude is the second quantisation $\Gamma(R)$ of the one-particle Bergman restriction $R: A^2(D_{\text{out}}) \rightarrow A^2(D_{\text{in},1}) \otimes A^2(D_{\text{in},2})$, with matrix elements satisfying*

$$|R_{n,m}| \leq Cq^{(n+m)/2},$$

where $q = e^{-t}$ and t is the conformal length of the collar. The sewing operator T_q on the reduced Bergman space $A_0^2(\partial D)$ has eigenvalues q^n for $n \geq 1$ and is trace class with

$$\|T_q\|_1 = \frac{q}{1-q}. \quad (4.2)$$

The genus- g partition function of $\mathcal{H}_k$ is the Fredholm determinant

$$Z_g(\mathcal{H}_k; \Omega) = (\det \text{Im } \Omega)^{-k/2} \cdot (\det'_\zeta \Delta_{\Sigma_g})^{-k/2}. \quad (4.3)$$

At genus 1, $Z_1 = (\text{Im } \tau)^{-k/2} |\eta(\tau)|^{-2k}$.

Proof. Wick's theorem: every Heisenberg amplitude factors into pairwise contractions of the generating current. Each contraction across a sewing circle corresponds to a one-particle pairing, so the full Fock-space amplitude equals the second quantisation $\Gamma(R)$ of the one-particle pairing R .

The one-particle matrix element is

$$R_{n,m} = \frac{k}{2\pi i} \oint_{|z|=1} \oint_{|w|=1} z^{-n-1} w^{-m-1} G_{\text{pants}}(z, w) dz dw,$$

where G_{pants} is the Bergman kernel of the pair of pants. The collar of length $t = -\log q$ and the Bergman reproducing property give $|G_{\text{pants}}(z, w)| \leq C(1 - q|zw|)^{-2}$; extracting the (n, m) Fourier coefficient yields $|R_{n,m}| \leq Cq^{(n+m)/2}$.

The operator R is Hilbert–Schmidt, so the second-quantised operator $\Gamma(R)$ is trace class, and $\|\Gamma(R)\|_1 = \det(1 + |R|)^k < \infty$.

The Polyakov–Belavin–Knizhnik formula [2] identifies the Heisenberg partition function: $\det(1 - K_g)^{-k} = (\det \text{Im } \Omega)^{-k/2} (\det'_\zeta \Delta_{\Sigma_g})^{-k/2}$, where K_g is the genus- g sewing kernel assembled by composing $2g - 2$ pair-of-pants restriction operators along the edges of a trivalent graph Γ of first Betti number g .

At genus 1, $K_1 = T_q$ and the Fredholm determinant is $\det(1 - T_q) = \prod_{n \geq 1} (1 - q^n) = q^{-1/24} \eta(q)$, so $Z_1 = (\operatorname{Im} \tau)^{-k/2} \cdot |\eta(\tau)|^{-2k}$ after combining holomorphic and antiholomorphic sectors. $\square$

Remark 4.5 (Genus-1 connected free energy). At genus 1, the connected free energy is

$$F_1^{\text{conn}}(\mathcal{H}_k; q) = -k \log \det(1 - T_q) = -k \log(q^{-1/24} \eta(q)) = k \left(\frac{1}{24} \log q - \sum_{n \geq 1} \log(1 - q^n) \right),$$

the Fourier expansion of which produces the connected sewing amplitudes $a_{\mathcal{H}_k}(N) = k \sum_{d|N} 1/d$; the Dirichlet series is $S_{\mathcal{H}_k}(u) = k \zeta(u) \zeta(u + 1)$. The Gram positivity of Corollary 3.4 is a consequence of $a_{\mathcal{H}_k}(N) > 0$.

4.6. Higher-genus Fredholm expansion. The genus- g partition function is assembled by choosing a pants decomposition of Σ_g : a trivalent graph Γ with g loops (first Betti number), $2g - 2$ vertices (pairs of pants), and $3g - 3$ internal edges (sewing circles). Each vertex contributes a pair-of-pants restriction operator; each edge contributes a sewing operator T_q (the restriction to a collar of conformal length $t = -\log q$).

The genus- g sewing kernel is the composition

$$K_g = \prod_{\text{edges of } \Gamma} T_{q_e} \circ \prod_{\text{vertices of } \Gamma} R_v, \quad (4.4)$$

where $q_e = e^{-t_e}$ is the sewing parameter of the e -th internal edge. The eigenvalues of K_g on the one-particle Bergman space are controlled by the Schottky parametrisation of Σ_g .

Proposition 4.6 (Eigenvalue asymptotics of the sewing kernel). *The eigenvalues λ_j of the genus- g sewing kernel K_g satisfy:*

- (i) $0 \leq \lambda_j < 1$ for all j ;
- (ii) $\sum_j \lambda_j < \infty$ (trace class);
- (iii) $\sum_j \lambda_j^2 < \infty$ (Hilbert–Schmidt);
- (iv) $\lambda_j \sim C_g j^{-1} \exp(-\alpha_g j^{1/g})$ as $j \rightarrow \infty$, where α_g depends on the conformal structure of Σ_g through the extremal length of the separating geodesic.

Proof. (i) and (ii): each T_{q_e} has eigenvalues q_e^n ($n \geq 1$), which are positive and summable. Composition of trace-class operators is trace class; eigenvalues of a composition are bounded by the product of spectral radii.

(iii): the square-summability follows from the Hilbert–Schmidt property of the pair-of-pants restriction (Theorem 4.4).

(iv): the asymptotic is determined by the extremal length of the shortest separating geodesic on Σ_g ; the exponential decay $\exp(-\alpha_g j^{1/g})$ reflects the g -fold nature of the Schottky parametrisation. $\square$

Corollary 4.7 (Fredholm expansion at genus g). *The genus- g partition function admits the absolutely convergent Fredholm expansion*

$$Z_g(\mathcal{H}_k; \Omega) = \det(1 - K_g)^{-k} = \exp\left(k \sum_{m \geq 1} \frac{1}{m} \text{Tr}(K_g^m)\right), \quad (4.5)$$

where the power traces $\text{Tr}(K_g^m)$ are convergent sums over products of sewing eigenvalues.

Proof. The Fredholm determinant of a trace-class operator satisfies $\log \det(1 - K) = -\sum_{m \geq 1} \frac{1}{m} \text{Tr}(K^m)$, convergent whenever $\|K\|_1 < 1$. The trace-class property of K_g (Proposition 4.6) ensures convergence; raising to the k -th power accounts for the level factor. $\square$

4.7. Genus-2 specialisation. At genus 2, every Riemann surface admits a hyperelliptic representation $y^2 = \prod_{i=1}^6 (x - e_i)$, and the period matrix is a 2×2 symmetric matrix $\Omega \in \mathfrak{H}_2$ with $\text{Im } \Omega > 0$. The Heisenberg partition function becomes

$$Z_2(\mathcal{H}_k; \Omega) = (\det \text{Im } \Omega)^{-k/2} \cdot (\det'_\zeta \Delta_{\Sigma_2})^{-k/2}. \quad (4.6)$$

The 2×2 determinant $\det \text{Im } \Omega = (\text{Im } \Omega_{11})(\text{Im } \Omega_{22}) - (\text{Im } \Omega_{12})^2$ is the volume of the Jacobian torus $\mathbb{C}^2/(\mathbb{Z}^2 + \Omega\mathbb{Z}^2)$.

The sewing kernel at genus 2 is assembled from a trivalent graph with 2 vertices and 3 internal edges: $K_2 = T_{q_1} \circ R_1 \circ T_{q_2} \circ R_2 \circ T_{q_3}$, where q_1, q_2, q_3 are the three sewing parameters corresponding to the three handles of the genus-2 surface (two non-separating, one separating). The degeneration limits $q_i \rightarrow 0$ recover the genus-1 partition functions of the constituent tori.

Remark 4.8 (Factorisation at the separating node). In the separating degeneration (one of the three sewing parameters approaches zero while the other two remain generic), the genus-2 partition function factors:

$$Z_2(\mathcal{H}_k; \Omega) \xrightarrow{q_3 \rightarrow 0} Z_1(\mathcal{H}_k; \tau_1) \cdot Z_1(\mathcal{H}_k; \tau_2) \cdot (1 + O(q_3)),$$

where τ_1, τ_2 are the modular parameters of the two genus-1 components. The leading term is the product of two genus-1 partition functions; the corrections are trace-class contributions from the sewing circle. The non-separating degeneration carries genuinely genus-2 information: the 2×2 period matrix controls the interplay of the two homology cycles.

Remark 4.9 (The Heisenberg as the atom of sewing). The Heisenberg algebra is the unique member of the standard landscape for which every step of the sewing programme is explicit: the sewing envelope is a classical Bergman space; the pair-of-pants amplitude is a Fock-space second quantisation; every partition function is a Fredholm determinant. All non-abelian chiral algebras satisfying HS-sewing require the abstract approach of Section 6. The lattice vertex algebra is the first extension (Section 5), where the Heisenberg Fredholm determinant appears as one factor of the full partition function.

4.8. The Heisenberg at higher genus: Schottky uniformisation. The Fredholm determinant formula for the Heisenberg partition function admits an interpretation in terms of Schottky uniformisation that illuminates the sewing structure.

A genus- g Schottky group is generated by g loxodromic Möbius transformations $\gamma_1, \dots, \gamma_g \in \mathrm{PSL}_2(\mathbb{C})$, each with multiplier q_j ($|q_j| < 1$). The Schottky uniformisation represents Σ_g as a quotient of a domain in $\mathbb{CP}^1$ by the Schottky group.

The sewing construction realises this uniformisation at the chain level: each generator γ_j corresponds to a sewing circle with parameter q_j , and the Fredholm determinant is

$$Z_g(\mathcal{H}_k; q_1, \dots, q_g) = \prod_{n \geq 1} \prod_{\{\gamma\}'} (1 - q_\gamma^n)^{-k}, \quad (4.7)$$

where $\{\gamma\}'$ denotes the set of primitive conjugacy classes in the Schottky group and q_γ is the multiplier of γ . The product converges absolutely because the multipliers decay exponentially: $|q_\gamma| \leq \max_j |q_j|^{| \gamma |}$, where $| \gamma |$ is the word length.

Remark 4.10 (Selberg zeta function). The Schottky product (4.7) is the inverse of the Selberg zeta function $Z_{\mathrm{Selberg}}(\Sigma_g, s)$ evaluated at $s = 1$. The Selberg zeta function encodes the spectrum of the Laplacian on Σ_g ; its value at $s = 1$ is $\det'_\zeta \Delta_{\Sigma_g}$. The Heisenberg partition function is therefore $Z_g(\mathcal{H}_k) = Z_{\mathrm{Selberg}}(\Sigma_g, 1)^{-k/2}$ times the period-matrix factor $(\det \mathrm{Im} \Omega)^{-k/2}$. This connects the algebraic sewing construction to spectral geometry.

Remark 4.11 (The Polyakov conjecture). Polyakov's programme computes the string partition function as a product of determinants: the holomorphic sector gives $(\det'_\zeta \Delta)^{-k/2}$ (the zeta-regularised determinant of the scalar Laplacian), and the non-holomorphic sector gives $(\det \mathrm{Im} \Omega)^{-k/2}$ (the volume of the Jacobian torus). The HS-sewing theorem provides the vertex-algebraic proof: the algebraic sewing (Fredholm determinant of the one-particle Bergman operator) reproduces the spectral-geometric answer (zeta-regularised determinant of the Laplacian), and both are equal to the Belavin–Knizhnik formula [2]. The three independent computations (algebraic, spectral, geometric) constitute a verification of the Heisenberg sewing theorem by independent paths.

5. LATTICE SEWING: THETA FUNCTIONS

The lattice vertex algebra V_Λ associated to a positive-definite even lattice Λ of rank r provides the natural next case after the Heisenberg: the sewing programme extends by combining the Heisenberg Fredholm determinant with the convergence of the Siegel theta function.

5.1. Structure of the lattice vertex algebra. Let Λ be a positive-definite even lattice of rank r , with dual lattice Λ^* and discriminant group $D(\Lambda) =$

Λ^*/Λ . The lattice vertex algebra V_Λ is the tensor product of the rank- r Heisenberg Fock space with the group algebra of Λ :

$$V_\Lambda = \bigoplus_{\gamma \in D(\Lambda)} \mathcal{F}_\gamma, \quad \mathcal{F}_\gamma = \mathcal{H}_r \otimes \mathbb{C}[[\Lambda + \gamma]],$$

where $\mathcal{F}_\gamma$ is the twisted Heisenberg module indexed by the coset $\gamma \in D(\Lambda)$. The vertex operators e^γ ($\gamma \in \Lambda$) act by

$$e^\gamma \cdot a_{-n_1} \cdots a_{-n_j} |\gamma'\rangle = \varepsilon(\gamma, \gamma') \cdot a_{-n_1} \cdots a_{-n_j} |\gamma + \gamma'\rangle,$$

where $\varepsilon: \Lambda \times \Lambda \rightarrow \{\pm 1\}$ is the 2-cocycle. The central charge is $c = r$ and $\kappa(V_\Lambda) = r$.

Sector dimensions satisfy $\dim(V_\Lambda)_n = |D(\Lambda)| \cdot p_r(n)$, where $p_r(n)$ is the r -coloured partition number. The sector growth is subexponential: $\log \dim(V_\Lambda)_n = O(\sqrt{n})$. OPE structure constants satisfy $|C_{a,i;b,j}^{c,k}| \leq K(a+b+c+1)^r$ because all OPE singularities in V_Λ are at most r -th order poles.

5.2. The lattice partition function.

Theorem 5.1 (Lattice sewing). *Let Λ be a positive-definite even lattice of rank r .*

- (i) (**HS-sewing**) V_Λ satisfies HS-sewing at every $0 < q < 1$.
- (ii) (**Factorisation**) The genus- g partition function factors as

$$Z_g(V_\Lambda; \Omega) = \underbrace{(\det \operatorname{Im} \Omega)^{-r/2} \cdot (\det'_\zeta \Delta_{\Sigma_g})^{-r/2}}_{Z_g(\mathcal{H}_r; \Omega)} \cdot \underbrace{\sum_{\lambda \in \Lambda^g} e^{i\pi \lambda^T \Omega \lambda}}_{\Theta_\Lambda(\Omega)}, \quad (5.1)$$

and each factor converges absolutely on the Siegel upper half-space $\mathfrak{H}_g$.

- (iii) (**Sewing envelope**) The sewing envelope is the charge-refined Hilbert completion

$$V_\Lambda^{\text{sew}} = \widehat{\bigoplus_{\gamma \in D(\Lambda)} \operatorname{Sym} A_\gamma^2(D)}, \quad (5.2)$$

where $A_\gamma^2(D) \subset A^2(D)$ is the Bergman space with L_0 -shift $\frac{1}{2}\langle \gamma, \gamma \rangle$ and the completion is in the HS-sewing topology.

- (iv) (**Bounded vertex operators**) The vertex operators $e^\gamma: \widehat{\mathcal{F}}_{\gamma'} \rightarrow \widehat{\mathcal{F}}_{\gamma'+\gamma}$ extend to bounded operators between completed sectors with $\|e^\gamma\|_{\text{op}} \leq q^{\langle \gamma, \gamma \rangle / 2}$.

Proof. (i) follows from $\log \dim(V_\Lambda)_n = O(\sqrt{n})$ (subexponential sector growth) and $|C_{a,i;b,j}^{c,k}| \leq K(a+b+c+1)^r$ (polynomial OPE growth): the hypotheses of Theorem 6.1 are satisfied.

(ii): the Fock decomposition $V_\Lambda = \bigoplus_\gamma \mathcal{F}_\gamma$ is a direct sum of twisted Heisenberg modules. Within each sector, Wick's theorem reduces all correlation functions to propagator pairings on the underlying rank- r Heisenberg, contributing $Z_g(\mathcal{H}_r; \Omega)$ (Theorem 4.4). The lattice sum over winding

modes contributes $\Theta_\Lambda(\Omega)$. The Heisenberg factor converges by the one-particle sewing theorem (eigenvalues q^n , trace class). The theta factor converges absolutely on $\mathfrak{H}_g$: the defining sum satisfies $|\Theta_\Lambda(\Omega)| \leq \sum_\lambda e^{-\pi \lambda^T Y \lambda}$ with $Y = \text{Im}(\Omega) > 0$, and the exponent $\lambda^T Y \lambda \geq \lambda_{\min}(Y) \cdot \|\lambda\|^2$ grows as $\|\lambda\|^2 \rightarrow \infty$.

(iii): the Bergman-space completion $\widehat{\mathcal{F}}_\gamma = \overline{\mathcal{F}_\gamma}^{\|\cdot\|_{A^2}}$ gives honest Hilbert sectors. The direct-sum decomposition of Proposition 2.4 applies to the $D(\Lambda)$ -grading.

(iv): the vertex operator e^γ preserves mode operators and only shifts the lattice vector. The cocycle ε is bounded by 1. The L_0 -shift $\Delta_\gamma = \frac{1}{2}\langle \gamma, \gamma \rangle$ is absorbed into the grading; the exponential weight damping q^{Δ_γ} in the HS topology gives the bound. $\square$

Remark 5.2 (Self-dual lattices and modular invariance). When Λ is unimodular ($D(\Lambda) = \{0\}$, e.g. E_8 or Λ_{24}), the theta function $\Theta_\Lambda(\Omega)$ is a Siegel modular form with respect to $\text{Sp}_{2g}(\mathbb{Z})$. The full partition function $Z_g(V_\Lambda; \Omega)$ then transforms as a modular form of weight $r/2$, recovering Zhu's modular invariance theorem [10] from the sewing formalism. For non-self-dual lattices, the theta function transforms under a finite-index congruence subgroup of $\text{Sp}_{2g}(\mathbb{Z})$, and the partition function has the corresponding transformation law.

Remark 5.3 (Genus-1 specialisation). At genus 1, the lattice partition function reduces to

$$Z_1(V_\Lambda; \tau) = \frac{\Theta_\Lambda(\tau)}{|\eta(\tau)|^{2r}}.$$

For the E_8 lattice ($r = 8$): the theta function $\Theta_{E_8}(\tau) = E_4(\tau)$ (the Eisenstein series of weight 4), and $Z_1(V_{E_8}; \tau) = E_4(\tau)/|\eta(\tau)|^{16}$. For the Leech lattice ($r = 24$): $\Theta_{\Lambda_{24}}(\tau) = E_{12}(\tau) + \dots$ (a weight-12 modular form).

Example 5.4 (Genus-2 for the E_8 lattice). At genus 2 with period matrix $\Omega = \begin{pmatrix} \tau_1 & z \\ z & \tau_2 \end{pmatrix} \in \mathfrak{H}_2$, the E_8 lattice partition function is

$$Z_2(V_{E_8}; \Omega) = Z_2(\mathcal{H}_8; \Omega) \cdot \Theta_{E_8}(\Omega),$$

where the Heisenberg factor is $Z_2(\mathcal{H}_8; \Omega) = (\det \text{Im } \Omega)^{-4} \cdot (\det'_\zeta \Delta_{\Sigma_2})^{-4}$ and the genus-2 theta function is

$$\Theta_{E_8}(\Omega) = \sum_{\lambda \in E_8^2} e^{i\pi \lambda^T \Omega \lambda} = \sum_{\lambda_1, \lambda_2 \in E_8} e^{i\pi(\tau_1 |\lambda_1|^2 + 2z \langle \lambda_1, \lambda_2 \rangle + \tau_2 |\lambda_2|^2)}.$$

The genus-2 theta function $\Theta_{E_8}(\Omega)$ is a Siegel modular form of weight 4 with respect to $\text{Sp}_4(\mathbb{Z})$. The factorisation of Z_2 into Heisenberg and lattice parts extends the genus-1 formula $Z_1 = E_4(\tau)/|\eta(\tau)|^{16}$ to genus 2.

The convergence of $\Theta_{E_8}(\Omega)$ on $\mathfrak{H}_2$ is controlled by the shortest vector of E_8 ($|\lambda_{\min}|^2 = 2$): the tail of the theta sum is bounded by $\sum_{|\lambda| \geq R} e^{-\pi \lambda_{\min}(Y) \cdot |\lambda|^2}$, which decays as $e^{-\pi \lambda_{\min}(Y) \cdot R^2}$.

Example 5.5 (The D_{16}^+ lattice and moonshine). The lattice D_{16}^+ (rank 16, even, unimodular, but not isomorphic to $E_8 \oplus E_8$) provides a lattice VOA with the same genus-1 partition function as $V_{E_8 \oplus E_8}$ (both equal $E_4(\tau)^2/|\eta(\tau)|^{32}$). The two VOAs are distinguished at genus 2: their Siegel theta functions $\Theta_{D_{16}^+}(\Omega) \neq \Theta_{E_8 \oplus E_8}(\Omega)$ are distinct Siegel modular forms of weight 8. The analytic sewing programme detects this distinction through the genus-2 partition function, providing finer invariants than the genus-1 character.

5.3. Convergence of the Siegel theta function. The theta factor $\Theta_\Lambda(\Omega) = \sum_{\lambda \in \Lambda^g} e^{i\pi \lambda^T \Omega \lambda}$ in the lattice partition function requires a separate convergence analysis from the Heisenberg Fredholm factor. The two convergence mechanisms are logically independent.

Proposition 5.6 (Theta convergence on the Siegel upper half-space). *Let Λ be a positive-definite even lattice of rank r . For any $\Omega \in \mathfrak{H}_g$ (i.e. $\Omega^T = \Omega$ and $Y := \text{Im } \Omega > 0$), the theta series converges absolutely with the bound*

$$|\Theta_\Lambda(\Omega)| \leq \sum_{\lambda \in \Lambda^g} e^{-\pi \lambda^T Y \lambda} \leq C(Y)^{r/2} \cdot (\det Y)^{-r/2}, \quad (5.3)$$

where $C(Y)$ is a constant depending on the lattice through the shortest vector.

Proof. For $\lambda = (\lambda_1, \dots, \lambda_g) \in \Lambda^g$, the real part of $i\pi \lambda^T \Omega \lambda$ is $-\pi \lambda^T Y \lambda$. Since $Y > 0$, each summand satisfies $|e^{i\pi \lambda^T \Omega \lambda}| = e^{-\pi \lambda^T Y \lambda}$. The quadratic form $\lambda^T Y \lambda \geq \lambda_{\min}(Y) \cdot \|\lambda\|^2$ grows as $\|\lambda\|^2$, so the sum converges by comparison with a Gaussian integral over $\mathbb{R}^{rg}$. The bound follows from the Poisson summation formula applied to the Gaussian $e^{-\pi \lambda^T Y \lambda}$. $\square$

Remark 5.7 (Two independent convergence mechanisms). The lattice partition function (5.1) has two factors, each with a distinct convergence mechanism:

- (a) the Heisenberg factor $Z_g(\mathcal{H}_r; \Omega)$ converges by the Fredholm determinant (trace-class sewing operator, eigenvalue decay q^n , $n \geq 1$);
- (b) the theta factor $\Theta_\Lambda(\Omega)$ converges by positive-definiteness of $\text{Im } \Omega$ (Gaussian decay of the lattice sum).

Neither mechanism alone suffices for the other factor. The Heisenberg factor involves the Bergman kernel and the Arakelov Green function; the theta factor involves the lattice geometry and the period matrix. The factorisation $Z_g = Z_g^{\text{Heis}} \cdot \Theta_\Lambda$ is a consequence of the Fock decomposition of V_Λ into twisted Heisenberg modules; no analogous factorisation is available for non-abelian algebras.

5.4. The Dirichlet–sewing lift for lattice VOAs. The connected free energy of the lattice VOA at genus 1 is

$$F_1^{\text{conn}}(V_\Lambda; q) = -r \log \det(1 - T_q) - \log \Theta_\Lambda(\tau), \quad (5.4)$$

where the first term is the rank- r Heisenberg contribution and the second is the lattice correction. The Dirichlet–sewing lift produces

$$S_{V_\Lambda}(u) = r \zeta(u) \zeta(u+1) + S_\Lambda^{\text{lattice}}(u), \quad (5.5)$$

where $S_\Lambda^{\text{lattice}}(u)$ encodes the theta function’s Fourier coefficients as a Dirichlet series. For self-dual lattices, the lattice Dirichlet series inherits the modular invariance of the theta function.

6. GENERAL HS-SEWING THEOREM FOR THE STANDARD LANDSCAPE

For chiral algebras beyond the Heisenberg and lattice families, the sewing envelope does not admit a classical description, but the Hilbert–Schmidt criterion can still be verified from two OPE-level estimates.

6.1. The general criterion.

Theorem 6.1 (General HS-sewing criterion). *Let $\mathcal{A}$ be a positive-energy chiral algebra with a choice of pre-Hilbert structure on each sector H_n . Suppose:*

- (i) **(subexponential sector growth)** $\log \dim H_n = o(n)$; equivalently, $\dim H_n \leq C e^{\alpha \sqrt{n}}$ for some $C, \alpha > 0$;
- (ii) **(polynomial OPE coefficient growth)** *there exist $K > 0$ and $N \geq 0$ such that the matrix elements of $m_{a,b}^c$ in orthonormal bases satisfy*

$$|C_{a,i;b,j}^{c,k}| \leq K(a+b+c+1)^N$$

for all weights a, b, c and all basis indices i, j, k .

Then $\mathcal{A}$ satisfies HS-sewing at every parameter $0 < q < 1$.

Proof. From hypothesis (ii), the Hilbert–Schmidt norm of the sector multiplication is bounded by

$$\|m_{a,b}^c\|_{\text{HS}}^2 \leq \dim H_a \cdot \dim H_b \cdot \dim H_c \cdot K^2(a+b+c+1)^{2N}. \quad (6.1)$$

The power-mean inequality $(x+y+z)^p \leq 3^{p-1}(x^p+y^p+z^p)$ with $p = 2N$ gives

$$(a+b+c+1)^{2N} \leq 3^{2N-1}((a+1)^{2N} + (b+1)^{2N} + (c+1)^{2N}).$$

Substituting into (6.1) and using the symmetry of the three terms:

$$\sum_{a,b,c \geq 0} q^{a+b+c} \|m_{a,b}^c\|_{\text{HS}}^2 \leq 3^{2N} K^2 \left(\sum_{n \geq 0} q^n \dim H_n (n+1)^{2N} \right) \left(\sum_{n \geq 0} q^n \dim H_n \right)^2.$$

By hypothesis (i), $\dim H_n \leq C e^{\alpha \sqrt{n}}$, while $q^n = e^{-n \log(1/q)}$ decays exponentially. Each factor $q^n \dim H_n (n+1)^{2N}$ is bounded by $C(n+1)^{2N} \exp(\alpha \sqrt{n} - n \log(1/q))$, which decays superpolynomially. Each series converges absolutely. $\square$

Corollary 6.2 (Automatic HS-sewing for Koszul algebras). *Every finitely strongly generated chirally Koszul algebra satisfies HS-sewing. The PBW theorem provides polynomial OPE growth; finite strong generation provides the finite N in hypothesis (ii); positive energy provides subexponential sector growth. HS-sewing is therefore an independent analytic condition only for infinitely generated algebras.*

6.2. Verification on the standard landscape. We verify the hypotheses of Theorem 6.1 for each standard family. The data required is a character formula (controlling $\dim H_n$) and a polynomial OPE bound (determining N).

Example 6.3 (Heisenberg). The Heisenberg $\mathcal{H}_k$ has $\dim H_n = p(n)$, the partition number. The OPE $J(z)J(w) \sim k/(z-w)^2$ has structure constants bounded by k , independent of weight; $N = 1$ suffices (the derivation ∂ increases weight by 1, so normal-ordered products of derivatives are at most linear in the weight).

Example 6.4 (Affine Kac–Moody). Let $V_k(\mathfrak{g})$ be the vacuum module at level k , with $\mathfrak{g}$ simple. Sectors satisfy $\dim H_n \leq Ce^{\alpha\sqrt{n}}$ from the Kac–Weyl character formula. The OPE $J^a(z)J^b(w) \sim k\delta^{ab}/(z-w)^2 + f^{ab}_c J^c(w)/(z-w)$ gives $|C| \leq K(n+1)$ with $N = 1$. HS-sewing at every $0 < q < 1$ follows.

Example 6.5 (Virasoro). The Virasoro algebra Vir_c has $\dim H_n = p(n)$. The OPE $T(z)T(w) \sim (c/2)(z-w)^{-4} + 2T(w)(z-w)^{-2} + \partial T(w)(z-w)^{-1}$ is quartic (the r -matrix has cubic pole by $d \log$ -absorption, per the pole relation $\text{pole}_r = \text{pole}_{\text{OPE}} - 1$). Repeated normal products give $|C| \leq K(n+1)^2$ with $N = 2$. The criterion applies at every central charge c .

Example 6.6 ($\mathcal{W}_N$ -algebras). The principal $\mathcal{W}_N$ -algebra has $N - 1$ generators of conformal weights $2, 3, \dots, N$. Sector dimensions satisfy $\dim H_n \leq Ce^{\alpha\sqrt{n}}$ from the $\mathcal{W}_N$ character formula. OPE coefficient growth is $|C| \leq K(n+1)^{2N}$; applying Theorem 6.1 gives HS-sewing at every $0 < q < 1$.

Example 6.7 (Lattice vertex algebras). Covered by Theorem 5.1; recorded here for completeness. $\dim H_n = |D(\Lambda)| \cdot p_r(n)$, $|C| \leq K \cdot r$, $N = r$.

Proposition 6.8 (Summary table). *The following table records the HS-sewing data for the standard landscape. Every entry satisfies the hypotheses of Theorem 6.1.*

$\mathcal{A}$	$\dim H_n$	$ C $ bound	N	$\sum_n q^n \dim H_n$
$\mathcal{H}_k$	$p(n)$	$\leq k$	1	$q^{1/24}/\eta(q)$
$V_k(\mathfrak{g})$	$\leq Ce^{\alpha\sqrt{n}}$	$\leq K(n+1)$	1	$\chi_{V_k}(q)$
Vir_c	$p(n)$	$\leq K(n+1)^2$	2	$q^{(1-c)/24}/\eta(q)$
$\mathcal{W}_N$	$\leq Ce^{\alpha\sqrt{n}}$	$\leq K(n+1)^{2N}$	$2N$	$\chi_{\mathcal{W}_N}(q)$
V_Λ	$ D(\Lambda) \cdot p_r(n)$	$\leq Kr$	r	$\Theta_\Lambda(q)/\eta(q)^r$

Remark 6.9 (Completed algebras). The standard families above are finitely strongly generated. For completed algebras (such as $\mathcal{W}_{1+\infty}$ or completed Yangians), the polynomial OPE bound of hypothesis (ii) must be verified in the completed (inverse-limit) topology, where the structure constants are formal power series. This verification is the content of the MC4 completion programme of [5].

Remark 6.10 (Beyond the standard landscape). The HS-sewing criterion is a priori applicable to any positive-energy chiral algebra satisfying subexponential growth and polynomial OPE bounds. Two classes of algebras beyond the standard landscape merit investigation:

- (a) Admissible-level simple quotients $L_k(\mathfrak{g})$ satisfy C_2 -cofiniteness [10] and hence HS-sewing (the C_2 -cofinite condition implies convergent characters, which implies subexponential growth; the simple quotient has finite-dimensional fused products, which implies polynomial OPE bounds).
- (b) $\mathcal{W}$ -algebras at non-generic central charge, where the sector growth may exhibit resonance phenomena but remains subexponential.

6.3. Fredholm expansion and the shadow tower. The HS-sewing theorem delivers convergent partition functions; the Fredholm expansion connects them to the shadow obstruction tower of the modular Koszul duality programme.

For any HS-sewing algebra, the genus- g partition function admits a graph-sum expansion over stable graphs $\Gamma \in \mathcal{S}_g$ (the set of stable graphs of $\overline{\mathcal{M}}_g$):

$$Z_g(\mathcal{A}; \Omega) = \sum_{\Gamma \in \mathcal{S}_g} \frac{1}{|\text{Aut}(\Gamma)|} \prod_{v \in V(\Gamma)} A_v(\mathcal{A}) \prod_{e \in E(\Gamma)} K_e(q_e), \quad (6.2)$$

where A_v is the vertex amplitude (genus- g_v with n_v legs) and $K_e(q_e)$ is the sewing kernel along edge e . Each term is absolutely convergent by the trace-class property.

The shadow obstruction tower $\Theta_{\mathcal{A}} = \sum_{r \geq 2} S_r(\mathcal{A})$ governs the leading-order asymptotics of the graph-sum. At uniform weight, the genus- g partition function satisfies $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$ (UNIFORM-WEIGHT), where λ_g^{FP} is the Faber–Pandharipande evaluation of the tautological class λ_g on $\overline{\mathcal{M}}_g$.

Proposition 6.11 (Convergence of the shadow tower evaluation). *Under HS-sewing, the evaluation of the MC element $\Theta_{\mathcal{A}}$ on any tautological class $\alpha \in H^*(\overline{\mathcal{M}}_{g,n})$ converges to a finite complex number. In particular, the genus- g shadow obstruction $F_g(\mathcal{A}) = \langle \Theta_{\mathcal{A}}, [\overline{\mathcal{M}}_g] \rangle$ is finite.*

Proof. The MC element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$ lies in the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$. Its evaluation on α is a sum of graph amplitudes $\langle \Theta_{\mathcal{A}}^{(r)}, \alpha \rangle$ indexed by the degree r of the MC component. Each graph amplitude factors through the trace-class sewing operators by the same mechanism as the partition function; the trace-class property ensures absolute convergence of each term and of the sum over r . $\square$

6.4. The Dirichlet–sewing lift. The connected free energy at genus 1 encodes connected sewing amplitudes $a_{\mathcal{A}}(N)$ as Fourier coefficients. The Dirichlet–sewing lift

$$S_{\mathcal{A}}(u) = \sum_{N \geq 1} a_{\mathcal{A}}(N) N^{-u} = \frac{(2\pi)^u}{\Gamma(u)} \int_0^\infty F_{\mathcal{A}}^{\text{conn}}(e^{-2\pi y}) y^{u-1} dy \quad (6.3)$$

is a Dirichlet series whose analytic structure is controlled by the MC element $\Theta_{\mathcal{A}}$.

Example 6.12 (Explicit Dirichlet series). The sewing Dirichlet series for the standard families:

$$S_{\mathcal{H}_k}(u) = k \zeta(u) \zeta(u+1), \quad (6.4)$$

$$S_{V_k(\mathfrak{g})}(u) = \dim(\mathfrak{g}) \cdot \zeta(u) \zeta(u+1), \quad (6.5)$$

$$S_{\text{Vir}_c}(u) = \zeta(u+1)(\zeta(u)-1), \quad (6.6)$$

$$S_{\mathcal{W}_N}(u) = \zeta(u+1) \left((N-1)\zeta(u) - \sum_{j=1}^{N-1} H_j(u) \right), \quad (6.7)$$

where $H_j(u) = \sum_{n=1}^j n^{-u}$ are partial harmonic sums. The Heisenberg and Kac–Moody series have a double pole at $u=0$; the Virasoro series has a simple pole (the subtraction of 1 from $\zeta(u)$ removes the $N=1$ term).

The sewing amplitudes $a_{\mathcal{A}}(N)$ define a positive-semidefinite inner product $\langle f, g \rangle_{\mathcal{A}} = \sum_N a_{\mathcal{A}}(N) f(N) g(N)$ with reproducing kernel $K_{\mathcal{A}}(s, t) = S_{\mathcal{A}}(s+t)$. The surface moment matrix $\mathcal{M}_{ij} = K_{\mathcal{A}}(\alpha/2 + i, \alpha/2 + j)$ is positive definite for all $\alpha \geq 2$: a Gram matrix $V^T D V$ with Vandermonde V .

7. THE THREE-LAYER ANALYTIC GAP

The HS-sewing theorem guarantees convergent partition functions for the standard landscape, but three layers of structure separate the convergence statement from a complete analytic theory.

7.1. Layer 1: the sewing envelope for interacting algebras. For the Heisenberg algebra, the sewing envelope is $\text{Sym} A^2(D)$: a classical Fock space built on the Bergman space of the disk. For lattice vertex algebras, it is the charge-refined Hilbert completion (5.2). Both identifications rest on the same mechanism: the Wick theorem reduces all amplitudes to one-particle (Heisenberg) or one-particle-plus-lattice (lattice) data, and the sewing envelope inherits the Bergman structure from the one-particle space.

For non-abelian chiral algebras (affine Kac–Moody at generic level, Virasoro, $\mathcal{W}_N$), no such reduction is available. The sewing envelope exists abstractly by Definition 2.1, and HS-sewing guarantees that the weighted completion H_q is a concrete Hilbert space on which all amplitudes are trace class. The open problem is a structural identification of $\mathcal{A}^{\text{sew}}$ analogous to the Bergman identification for the Heisenberg.

Conjecture 7.1 (Sewing envelope for $V_k(\mathfrak{g})$). *The sewing envelope of $V_k(\mathfrak{g})$ at integer level $k \in \mathbb{Z}_{>0}$ is the ind-Hilbert completion $\widehat{V_k(\mathfrak{g})}^{\text{IndHilb}}$, and the pair-of-pants amplitude admits a Fredholm expansion indexed by Weyl-orbit data.*

The next step is explicit: the case $\mathfrak{g} = \mathfrak{sl}_2$ at $k = 1$ is the simplest interacting sewing computation.

Example 7.2 ($\mathfrak{sl}_2$ at level 1). The vacuum module $V_1(\mathfrak{sl}_2)$ has central charge $c = 1$ and $\kappa = \dim(\mathfrak{sl}_2)(1 + 2)/(2 \cdot 2) = 3 \cdot 3/4 = 9/4$. At conformal weight 2, the sector H_2 is 3-dimensional, spanned by $\{J_{-1}^a J_{-1}^b \mathbf{1}\}_{a,b \in \{+, -, 0\}}$ modulo the Serre relations. The pair-of-pants sewing kernel restricted to this sector is a 3×3 matrix whose entries are determined by the fusion rules $V_1 \otimes V_1 \rightarrow V_1$ and the KZ connection at level 1.

The KZ equation at level 1 is $\partial_z \Phi = \frac{\Omega}{(1+2)z} \Phi = \frac{\Omega}{3z} \Phi$ (KZ normalisation with Sugawara denominator $k + h^\vee = 3$), where Ω is the Casimir in $\mathfrak{sl}_2 \otimes \mathfrak{sl}_2$. The eigenvalues of the sewing kernel K_1 on the 3-dimensional sector are $\{q, q^2, q^2\}$ (the level-1 WZW primary fields have conformal weights 0 and $1/4$), so the trace-class condition is satisfied with $\|K_1|_{H_2}\|_1 = q + 2q^2$.

This is the gateway to a full interacting sewing computation: the first case where the sewing kernel is a non-diagonal matrix and the Fredholm expansion involves genuine fusion data.

Remark 7.3 (The conformal-block obstruction). For non-abelian chiral algebras at generic (non-integer) level, the spaces of conformal blocks are infinite-dimensional; the sewing kernel is an infinite-rank operator on these spaces. HS-sewing guarantees that this operator is trace class, but the explicit spectral decomposition requires detailed knowledge of the conformal blocks (solutions of the KZ equation). For integer level, the conformal blocks are finite-dimensional (Zhu's theorem), and the sewing kernel is a finite matrix at each degree; this is the regime where explicit computation is feasible.

7.2. Layer 2: metric independence of the ind-Hilbert factorisation.

Moriwaki's ind-Hilbert completion [6] is defined using a conformal metric on the curve; the Bergman inner product on $A^2(D)$ depends on the disk D and hence on the conformal structure. The algebraic bar complex, by contrast, depends only on the OPE data and is independent of any metric.

Definition 7.4 (Metric independence). An ind-Hilbert factorisation is *metrically independent* if the isomorphism class of the completed sewing envelope (as a topological vector space with continuous pair-of-pants multiplication) is independent of the conformal metric used to define the Bergman inner product.

For the Heisenberg, metric independence is immediate: the Fock space $\text{Sym} A^2(D)$ depends on the disk D , but any two disks are conformally equivalent, and the isomorphism class of the Bergman space is a conformal invariant.

For interacting algebras, metric independence is the assertion that the analytic completion sees only the algebraic structure (the OPE), not the auxiliary metric. This is a functional-analytic statement with no algebraic counterpart; it is the content of the algebraic-to-analytic comparison theorem.

Conjecture 7.5 (Metric independence). *For every positive-energy chiral algebra satisfying HS-sewing, the ind-Hilbert factorisation is metrically independent: the sewing envelope $\mathcal{A}^{\text{sew}}$ depends only on the algebraic chiral core $\mathcal{A}_{\text{alg}}$.*

Remark 7.6 (Evidence for metric independence). Three observations support the conjecture:

- (a) For the Heisenberg, the Bergman space $A^2(D)$ depends on D , but the abstract Hilbert space structure (separable, infinite-dimensional, with the canonical grading by polynomial degree) is a conformal invariant. Any two conformal metrics give isomorphic Bergman spaces; the isomorphism is the conformal change-of-variables on holomorphic functions.
- (b) The algebraic bar differential depends only on OPE data (residues at collision diagonals); the OPE coefficients are independent of conformal metric. If the bar differential is dense in the completed bar differential (as guaranteed by the analytic-algebraic comparison), the topology of the completion is determined by the algebraic data.
- (c) The sewing envelope is defined by seminorms (2.1) that depend on bordism amplitudes. Each amplitude is a composition of OPE operations (metric-independent) and sewing traces (metric-dependent only through the choice of conformal collar). The collar dependence enters through the sewing parameter q , which appears universally in all completions.

The conjecture is that these observations extend to a proof for all HS-sewing algebras: the sewing topology depends on q but not on the conformal metric that determines the embedding of the collar in the surface.

7.3. Layer 3: the coderived shadow at $g \geq 1$. At genus $g = 0$ the bar complex satisfies $d^2 = 0$, and the ordinary derived category is the correct receptacle. At genus $g \geq 1$, the fiberwise bar differential acquires curvature:

$$d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g, \quad (7.1)$$

where $\kappa(\mathcal{A})$ is the modular characteristic and $\omega_g = c_1(\lambda)$ is the first Chern class of the Hodge bundle on $\overline{\mathcal{M}}_g$. For $\kappa \neq 0$ (which holds for every non-trivial chiral algebra in the standard landscape), $d_{\text{fib}}^2 \neq 0$ and every object is acyclic in the ordinary derived category.

The correct receptacle is the coderived category D^{co} , developed in the next section. The passage from derived to coderived is the categorified form of passage from the equation $d^2 = 0$ to the curved Maurer–Cartan equation $d\alpha + \alpha^2 + \kappa\omega = 0$.

The analytic content of Layer 3 is: the sewing amplitudes that converge by HS-sewing must be reinterpreted as evaluations of coderived (not derived) functors. The Fredholm determinant of the Heisenberg is the prototype: the curvature $d_{\text{fib}}^2 = k \cdot \omega_1$ at genus 1 means that the bar complex does not define an object in the derived category; the Fredholm determinant $\det(1 - T_q) = q^{-1/24} \eta(q)$ is the correct coacyclic invariant.

7.4. Interaction between the three layers. The three layers are logically independent but interact in concrete examples.

Proposition 7.7 (Layer independence).

- (i) *Layer 1 (sewing envelope identification) does not imply Layer 2 (metric independence): an explicit Hilbert identification could depend on the choice of conformal metric.*
- (ii) *Layer 2 (metric independence) does not imply Layer 3 (coderived formulation): metric independence is a genus-0 property of the completion, while the coderived category is forced by curvature at $g \geq 1$.*
- (iii) *Layer 3 (coderived formulation) does not imply Layer 1: the coderived category provides the correct homotopy theory but does not identify the completion concretely.*

Proof. (i): the Heisenberg sewing envelope $\text{Sym}A^2(D)$ depends on the disk D ; the metric independence is a separate theorem (conformal invariance of the Bergman space, Proposition A.6).

(ii): metric independence concerns the topology of the completion; the coderived category concerns the homological algebra of the curved differential. The Heisenberg at genus 0 is metrically independent but has $d^2 = 0$ (no coderived issues); at genus 1 the coderived formulation is needed but the metric independence is inherited from genus 0.

(iii): the coderived category provides $C_{\text{BV}}^\bullet \simeq_{D^\infty} B^{(g)}$ abstractly; the identification of $\mathcal{A}^{\text{sew}}$ with a specific Hilbert space is additional data. $\square$

For the standard landscape, the current status of each layer:

Family	Layer 1	Layer 2	Layer 3
Heisenberg	proved	proved	proved
Lattice	proved	proved	proved
Affine KM	conjectural	conjectural	proved
Virasoro	conjectural	conjectural	proved
$\mathcal{W}_N$	conjectural	conjectural	proved
$\beta\gamma$	conjectural	conjectural	conditional

Layer 3 is proved for all standard families (Theorem 8.7); Layers 1 and 2 are proved only for the abelian families (Heisenberg and lattice). The resolution of Layers 1 and 2 for non-abelian families is the primary open problem of the analytic sewing programme.

7.5. The sewing envelope as a nuclear space. The sewing envelope carries a natural nuclear topology (projective limit of Hilbert spaces H_q as $q \rightarrow 1^-$). This nuclear structure provides additional functional-analytic tools.

Proposition 7.8 (Nuclear structure). *Under HS-sewing, the sewing envelope $\mathcal{A}^{\text{sew}}$ is the projective limit*

$$\mathcal{A}^{\text{sew}} = \varprojlim_{q \rightarrow 1^-} H_q,$$

and the transition maps $H_{q'} \rightarrow H_q$ (for $q' > q$) are Hilbert–Schmidt. In particular, $\mathcal{A}^{\text{sew}}$ is a nuclear Fréchet space.

Proof. The weighted completions H_q form a projective system: for $q' > q$, the inclusion $\mathcal{A}_{\text{alg}} \hookrightarrow H_q$ factors through $H_{q'}$ because $q'^n > q^n$ for all $n \geq 1$. The transition map $H_{q'} \rightarrow H_q$ sends the basis vector $e_{n,j} \in H_n$ to $(q/q')^n e_{n,j}$, with Hilbert–Schmidt norm $\sum_{n,j} (q/q')^{2n} = \sum_n \dim H_n (q/q')^{2n}$, which is finite by subexponential sector growth (the ratio $q/q' < 1$ provides exponential decay). A projective limit of Hilbert spaces with Hilbert–Schmidt transition maps is nuclear [8]. $\square$

The nuclear structure implies the Schwartz kernel theorem: every continuous bilinear form on $\mathcal{A}^{\text{sew}}$ is represented by a distribution kernel. This provides the correct framework for the pair-of-pants multiplication as a continuous bilinear operation on the sewing envelope.

8. THE CODERIVED CATEGORY AS CORRECT RECEPTACLE

The genus $g \geq 1$ bar complex carries curvature $d_{\text{fb}}^2 = \kappa \cdot \omega_g$. The ordinary derived category is empty: every curved complex is acyclic. The coderived category D^{co} resolves this: curved differentials are permitted, coacyclic complexes replace acyclic ones, and the BV/bar comparison holds universally.

8.1. Curved dg-modules.

Definition 8.1 (Curved dg-module). A *curved dg-module* over a graded algebra R is a graded R -module M equipped with a degree-1 endomorphism d satisfying

$$d^2 = \mu \cdot \text{id}_M$$

for some central element $\mu \in R$ (the *curvature*). When $\mu = 0$ this reduces to an ordinary dg-module.

The genus- g bar complex $B^{(g)}(\mathcal{A})$ is a curved dg-module over $R = H^*(\overline{\mathcal{M}}_g)$ with curvature $\mu = \kappa(\mathcal{A}) \cdot \omega_g$.

Warning 8.2 (Acyclicity in the ordinary sense). If $d^2 = \mu \neq 0$, then for every cocycle x ($dx = 0$) we have $\mu \cdot x = d^2x = d(dx) = 0$; if μ is not a zero divisor, this forces $x = 0$. In particular, $H^*(M, d) = 0$ whenever μ is

a non-zero-divisor. The ordinary derived category of curved dg-modules is therefore trivial when $\kappa \neq 0$.

8.2. The coderived category. The coderived category replaces acyclicity with coacyclicity.

Definition 8.3 (Coacyclic complex). A curved dg-module M is *coacyclic* if it lies in the smallest thick subcategory of the homotopy category of curved dg-modules containing all totalizations of short exact sequences. Equivalently, M is coacyclic if every morphism from M to an injective curved dg-module is null-homotopic.

Definition 8.4 (Coderived category). The *coderived category* $D^{\text{co}}(R\text{-mod}_\mu)$ is the Verdier quotient of the homotopy category of curved dg-modules by the thick subcategory of coacyclic complexes.

The coderived category is the correct homotopy category for curved complexes: it retains the meaningful homological information that the ordinary derived category discards.

Remark 8.5 (Positselski's theory). The coderived category was introduced by Positselski [7] in the context of curved dg-algebras and matrix factorisations. The totalisation of a short exact sequence of curved complexes need not be acyclic (because $d^2 \neq 0$), but it is always coacyclic: coacyclicity is the correct generalisation of acyclicity to the curved setting.

8.3. The curvature bicomplex. The curved bar complex $B^{(g)}(\mathcal{A})$ carries a natural bicomplex structure. Write the total differential as

$$D_g = d_{\text{fib}} + \nabla_g^{\text{GM}},$$

where d_{fib} is the fiberwise bar differential (with curvature $d_{\text{fib}}^2 = \kappa \cdot \omega_g$) and ∇_g^{GM} is the Gauss–Manin connection. The total differential satisfies

$$D_g^2 = d_{\text{fib}}^2 + [d_{\text{fib}}, \nabla_g^{\text{GM}}] + (\nabla_g^{\text{GM}})^2 = \kappa\omega_g - \kappa\omega_g = 0 : \quad (8.1)$$

flatness is restored by coupling the bar differential to the Hodge bundle. The curvature bicomplex is the bigraded object with horizontal differential d_{fib} and vertical differential ∇_g^{GM} .

Proposition 8.6 (Totalisation in D^{co}). *The totalisation of the curvature bicomplex of $B^{(g)}(\mathcal{A})$ is an object of the coderived category D^{co} . When $\mathcal{A}$ satisfies HS-sewing, the associated partition function is the trace of the totalised sewing operator and coincides with the absolutely convergent Fredholm expansion of Corollary 3.3.*

Proof. The totalisation of the bicomplex $(d_{\text{fib}}, \nabla_g^{\text{GM}})$ produces a complex with $D_g^2 = 0$ by (8.1), hence an honest object of the derived category of the total complex. The passage to the coderived category of the curved component d_{fib} (with $d_{\text{fib}}^2 = \kappa\omega_g$) is by the standard Positselski comparison:

totalisation of the bicomplex represents the curved component as an object of D^{co} because the vertical differential ∇_g^{GM} absorbs the curvature.

Under HS-sewing, the trace-class property of Proposition 3.2 ensures that the trace of the totalised sewing operator is an absolutely convergent number; this trace coincides with the partition function computed from the Fredholm expansion, because both are the same composition of pair-of-pants operators evaluated on the completed Hilbert space H_q . $\square$

8.4. The BV/bar comparison in the coderived category. The BV complex of a chiral algebra on a Riemann surface Σ_g is the Costello–Gwilliam perturbative quantisation [3]; the bar complex $B^{(g)}(\mathcal{A})$ is the algebraic chain model. At genus 0, the two are chain-level quasi-isomorphic. At genus $g \geq 1$, curvature intervenes: the BV propagator acquires a harmonic component (the Arakelov Green function) that produces corrections invisible to the algebraic bar differential.

Theorem 8.7 (BV = bar in the coderived category). *Let $\mathcal{A}$ be a chirally Koszul algebra. For each genus $g \geq 1$, let*

$$f_g : C_{\text{BV}}^\bullet(\mathcal{A}, \Sigma_g) \longrightarrow B^{(g)}(\mathcal{A})$$

be the comparison morphism obtained by replacing the BV propagator with its Hodge decomposition relative to the bar propagator.

(i) *For every genus $g \geq 0$,*

$$C_{\text{BV}}^\bullet(\mathcal{A}, \Sigma_g) \simeq_{D^{\text{co}}} B^{(g)}(\mathcal{A}).$$

- (ii) *If $\mathcal{A}$ lies in class G or L, then f_g is a weak equivalence of filtered curved models.*
- (iii) *If $\mathcal{A}$ lies in class C and harmonic decoupling holds, then f_g is a weak equivalence.*
- (iv) *If $\mathcal{A}$ lies in class M, then for every $r \geq 4$ the harmonic discrepancy satisfies*

$$\delta_r^{\text{harm}} = c_r(\mathcal{A}) \cdot m_0^{\lfloor r/2 \rfloor - 1}$$

and the cone of f_g is coacyclic.

Proof sketch. At genus 0, the comparison is the chain-level quasi-isomorphism: $d_{\text{fib}}^2 = 0$ and the harmonic component of the propagator vanishes on $\mathbb{C}$.

For $g \geq 1$, Hodge decomposition writes the BV propagator as the sum of the bar propagator (holomorphic, $d \log E(z, w)$, weight 1) and the harmonic correction $h_g = (2\pi)^{-1} (\text{Im } \Omega)_{ab}^{-1} d\bar{z}_a \wedge dz_b$. The harmonic correction produces a discrepancy between the BV and bar differentials.

For classes G and L (Heisenberg, affine Kac–Moody): the harmonic correction to the cubic vertex vanishes by Jacobi (for L) or by absence of interaction vertices (for G). The comparison is a weak equivalence of filtered curved models.

For class C ($\beta\gamma$): composite free-field factorisation reduces the harmonic correction to a product of Heisenberg corrections; each factor vanishes by the G-class argument, conditional on harmonic decoupling.

For class M (Virasoro, $\mathcal{W}_N$): the quartic harmonic discrepancy $\delta_4^{\text{harm}} \propto Q^{\text{contact}} \cdot \kappa / \text{Im}(\tau)$ is not cancelled by the Fay trisecant identity. The naive chain-level comparison fails. The full harmonic discrepancy is a sum of curvature powers $\delta_r^{\text{harm}} = c_r(\mathcal{A}) \cdot m_0^{\lfloor r/2 \rfloor - 1}$, and the comparison cone is coacyclic in the sense of Definition 8.3: every morphism from the cone to an injective curved module is null-homotopic, because the discrepancy is divisible by the curvature element m_0 .

The coderived equivalence (i) follows: the cone of f_g is coacyclic for every class, and a morphism with coacyclic cone is an isomorphism in D^{co} . $\square$

Remark 8.8 (Class-by-class obstruction profile). The four shadow classes of the standard landscape exhibit a strict hierarchy of analytic regularity:

Class	Chain-level status	Mechanism
G (Heisenberg)	weak equivalence	no interaction vertices
L (affine KM)	weak equivalence	Jacobi kills cubic harmonic term
C ($\beta\gamma$)	conditional	composite free-field factorisation
M (Virasoro, $\mathcal{W}_N$)	coderived only	harmonic correction curvature-divisible

For classes G and L the chain-level and coderived comparisons coincide. For class M the chain-level comparison genuinely fails, and the coderived category is essential.

8.5. The harmonic discrepancy mechanism. The key to the coderived comparison is the factorisation of the harmonic discrepancy through curvature powers. The mechanism is the Hodge decomposition of the Bergman kernel.

On a genus- g surface Σ_g with period matrix Ω , the Bergman kernel (the bar propagator) is

$$B(z, w) = d_z d_w \log E(z, w) + 2\pi i (\text{Im } \Omega)_{ab}^{-1} \omega_a(z) \omega_b(w), \quad (8.2)$$

where $E(z, w)$ is the prime form and $\omega_1, \dots, \omega_g$ is a normalised basis of abelian differentials. The second term is the harmonic correction.

The BV propagator P_g differs from the bar propagator $d \log E(z, w)$ by this harmonic term:

$$P_g(z, w) = d \log E(z, w) + h_g(z, w), \quad h_g = \frac{1}{2\pi} (\text{Im } \Omega)_{ab}^{-1} d\bar{z}_a \wedge dz_b.$$

Each insertion of h_g into a Feynman graph vertex produces a harmonic correction to the BV amplitude. At the quartic level ($r = 4$), the correction is

$$\delta_4^{\text{harm}} = Q^{\text{contact}}(\mathcal{A}) \cdot \frac{\kappa(\mathcal{A})}{\text{Im}(\tau)}, \quad (8.3)$$

where Q^{contact} is the quartic contact invariant of $\mathcal{A}$. For class G and L, $Q^{\text{contact}} = 0$. For class M, $Q^{\text{contact}} \neq 0$ and δ_4^{harm} is a genuine obstruction to the chain-level comparison.

The crucial observation is that every δ_r^{harm} is divisible by $m_0 = \kappa \cdot \omega_g$ (the curvature element of the curved bar differential). This divisibility is the mechanism by which the comparison cone becomes coacyclic: in the coderived category, elements divisible by the curvature are trivial.

8.6. Coderived versus derived: a comparison. The passage from derived to coderived captures a precise mathematical phenomenon.

	Derived $D(R)$	Coderived $D^{\text{co}}(R)$
Curvature	$d^2 = 0$ required	$d^2 = \mu$ permitted
Acyclicity notion	$H^*(M) = 0$	coacyclic
Genus-0 bar	object of $D(R)$	object of $D^{\text{co}}(R)$
Genus- g bar ($\kappa \neq 0$)	empty	well-defined
BV/bar comparison	chain-level (G,L)	all classes
Partition function	formal series	convergent number

The coderived category is the categorified avatar of the curved Maurer–Cartan equation. The passage from $d^2 = 0$ to $d\alpha + \alpha^2 + \kappa\omega = 0$ is the passage from derived to coderived: the curvature becomes data rather than obstruction.

8.7. Genus-1 illustration. The genus-1 case illustrates the three categories concretely.

At genus 1, the bar complex $B^{(1)}(\mathcal{A})$ has curvature $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_1$, where $\omega_1 = c_1(\lambda)$ is the first Chern class of the Hodge bundle on $\overline{\mathcal{M}}_{1,1}$. The total differential $D_1 = d_{\text{fib}} + \nabla^{\text{GM}}$ satisfies $D_1^2 = 0$.

Example 8.9 (Heisenberg at genus 1). For $\mathcal{H}_k$ with $\kappa = k$:

- (a) *Derived category:* $d_{\text{fib}}^2 = k \cdot \omega_1 \neq 0$ for $k \neq 0$. Every object in the curved complex is acyclic: $H^*(B^{(1)}(\mathcal{H}_k), d_{\text{fib}}) = 0$.
- (b) *Coderived category:* the totalisation of the bicomplex $(d_{\text{fib}}, \nabla^{\text{GM}})$ is a well-defined object of D^{co} . The trace of the sewing operator gives $Z_1 = (\text{Im } \tau)^{-k/2} |\eta(\tau)|^{-2k} \neq 0$.
- (c) *Total category:* $D_1^2 = 0$; the total complex $B^{(1)}(\mathcal{H}_k)$ is an honest dg-module over $H^*(\overline{\mathcal{M}}_{1,1})$ and lives in the ordinary derived category of the total complex.

The partition function Z_1 is a coderived invariant (well-defined in D^{co} of the curved fiberwise complex) and simultaneously a derived invariant (well-defined in D of the total complex). The two perspectives are related by the totalisation functor of Proposition 8.6.

Example 8.10 (Virasoro at genus 1). For Vir_c with $\kappa = c/2$:

- (a) The curvature is $d_{\text{fib}}^2 = (c/2) \cdot \omega_1$.

- (b) The partition function is $Z_1(\text{Vir}_c; \tau) = (\text{Im } \tau)^{-c/4} |\eta(\tau)|^{-2(c-1)}$ (the vacuum character of the Virasoro algebra).
- (c) The BV/bar comparison: the harmonic discrepancy at the quartic level is $\delta_4^{\text{harm}} = Q^{\text{contact}} \cdot (c/2)/\text{Im}(\tau)$. For $c \neq 0$ (non-trivial Virasoro), this is nonzero; the chain-level BV/bar comparison fails. The cone is coacyclic by Theorem 8.7(iv), so the coderived comparison holds.

The Virasoro at genus 1 is the simplest class-M example: the harmonic discrepancy is explicit, the coacyclicity is verifiable, and the coderived category is genuinely necessary.

8.8. Convergent versus formal partition functions. The three levels of structure (algebraic, coderived, analytic) produce three notions of partition function:

- (i) *Formal*: the genus- g component of the MC element $\Theta_{\mathcal{A}}$, evaluated on the fundamental class $[\overline{\mathcal{M}}_g]$. This is a formal power series in the sewing parameters $q_1, \dots, q_g$. It exists for every chiral algebra with $D^2 = 0$.
- (ii) *Coderived*: the trace of the totalised sewing operator in the coderived category. This is a formal power series with the additional property that each coefficient is a coderived invariant (independent of the choice of resolution up to coacyclic error). It exists for every positive-energy chiral algebra.
- (iii) *Analytic*: an absolutely convergent complex number, the trace of a trace-class operator on the weighted Hilbert completion H_q . This exists under HS-sewing.

Proposition 8.11 (Compatibility of the three notions). *For a chirally Koszul algebra satisfying HS-sewing:*

- (a) *the formal partition function is the Taylor expansion of the analytic partition function around the cusp $q_j \rightarrow 0$;*
- (b) *the coderived partition function is the image of the analytic partition function under the forgetful functor from the Hilbert-completed category to the coderived category;*
- (c) *all three notions agree on the domain of convergence.*

Proof. (a): the formal MC element $\Theta_{\mathcal{A}}$ is the generating function for the graph-sum amplitudes, which are the Taylor coefficients of the partition function at the cusp. Under HS-sewing, the Taylor series converges absolutely for $|q_j| < 1$ (Corollary 3.3), so the formal and analytic partition functions agree on the domain of convergence.

(b): the forgetful functor sends a trace-class operator on H_q to its underlying curved dg-module structure in D^{co} ; the trace (a convergent number) is sent to the same number viewed as a coderived invariant.

(c): on the domain of convergence, all three are the same complex number, computed by the same graph-sum with the same absolutely convergent Fredholm expansion. $\square$

8.9. The analytic Koszul pair. The algebraic Koszul pair $(\mathcal{A}, \mathcal{A}^!)$ controls Theorems A through H of the modular Koszul duality programme at the algebraic level. The analytic extension is the following.

Definition 8.12 (Analytic Koszul pair). An *analytic Koszul pair* is a pair $(\mathcal{A}, \mathcal{A}^!)^{\text{an}}$ of chirally Koszul algebras, both satisfying HS-sewing, for which the analytic bar-cobar adjunction restricts to a Quillen equivalence on sewing-complete categories:

$$\bar{B}^{\text{an}} : \mathcal{A}^{\text{sew}}\text{-mod} \rightleftarrows (\mathcal{A}^!)^{\text{sew}}\text{-comod} : \Omega^{\text{an}}.$$

Every algebraic Koszul pair in the standard landscape is expected to extend to an analytic Koszul pair; the HS-sewing theorem provides the analytic input for both $\mathcal{A}$ and $\mathcal{A}^!$.

Conjecture 8.13 (Analytic realisation criterion). *HS-sewing on $\mathcal{A}$ plus dual HS-sewing on $\mathcal{A}^!$ suffices for the analytic Koszul pair to exist: the analytic bar-cobar adjunction restricts to a Quillen equivalence on sewing-complete categories.*

The verification of this conjecture for the Heisenberg ($\mathcal{H}_k^! = \text{Sym}^{\text{ch}}(V^*) \neq \mathcal{H}_{-k}$) and for lattice vertex algebras is the natural test case.

Remark 8.14 (Koszul complementarity and the analytic pair). The Koszul complementarity theorem (Theorem C of the modular Koszul duality programme) asserts that $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = \varkappa(\mathcal{A})$ (the scalar complementarity invariant): $\varkappa = 0$ for KM/Heis/lattice/free; $\varkappa = 13$ for Virasoro; $\varkappa = 250/3$ for $\mathcal{W}_3$. The scalar $\varkappa$ is distinct from the Trinity ghost-charge conductor $K(\mathcal{A}) := c(\mathcal{A}) + c(\mathcal{A}^!) = -c_{\text{ghost}}()$ (taking values $K(\text{Vir}) = 26$, $K(\mathcal{W}_3) = 100$, $K(\text{BP}) = 196$); the two invariants are related by $\varkappa = \varrho_{\mathcal{A}} \cdot K$ with ϱ family-specific. The analytic content is: both $\mathcal{A}$ and $\mathcal{A}^!$ satisfy HS-sewing (because both are in the standard landscape), and the complementarity relation constrains the sewing data:

$$S_{\mathcal{A}}(u) + S_{\mathcal{A}^!}(u) = \varkappa \cdot \zeta(u) \zeta(u+1) + \text{lower-order terms}, \quad (8.4)$$

where the leading term involves the scalar complementarity invariant. For the Virasoro pair $(\text{Vir}_c, \text{Vir}_{26-c})$: $S_{\text{Vir}_c}(u) + S_{\text{Vir}_{26-c}}(u) = 2\zeta(u+1)(\zeta(u)-1)$, independent of c ; the dependence on c cancels in the sum, reflecting the topological nature of the Koszul conductor.

Remark 8.15 (The analytic bar coalgebra). The algebraic bar coalgebra $B(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ carries the deconcatenation coproduct $\Delta : B(\mathcal{A}) \rightarrow B(\mathcal{A}) \otimes B(\mathcal{A})$. Under HS-sewing, the bar coalgebra completes to the *analytic bar coalgebra* $B^{\text{an}}(\mathcal{A})$, defined as the graph-norm closure of $B(\mathcal{A})$ in the Hilbert-space topology induced by H_q . The coproduct extends continuously:

$\Delta^{\text{an}}: B^{\text{an}}(\mathcal{A}) \rightarrow B^{\text{an}}(\mathcal{A}) \widehat{\otimes} B^{\text{an}}(\mathcal{A})$, where $\widehat{\otimes}$ is the completed Hilbert-space tensor product.

The analytic bar coalgebra is the correct object for the analytic bar-cobar adjunction: it carries both the algebraic coalgebra structure (deconcatenation) and the analytic topology (trace-class sewing), and the cobar functor Ω^{an} applied to $B^{\text{an}}(\mathcal{A})$ recovers the sewing envelope $\mathcal{A}^{\text{sew}}$.

8.10. The curvature–sewing interplay. The coderived category and the HS-sewing condition interact at genus $g \geq 1$ through the curvature.

Proposition 8.16 (Curvature and trace class). *Let $\mathcal{A}$ satisfy HS-sewing and let $\kappa = \kappa(\mathcal{A})$. At genus $g \geq 1$, the curvature element $m_0 = \kappa \cdot \omega_g$ acts on the weighted completion H_q as a bounded operator of norm $\|m_0\|_{\text{op}} = |\kappa| \cdot \|\omega_g\|$, where $\|\omega_g\|$ is the sup-norm of the Hodge class representative on the moduli space. The iterated curvatures m_0^k remain bounded for all k , and the harmonic discrepancy $\delta_r^{\text{harm}} = c_r \cdot m_0^{\lfloor r/2 \rfloor - 1}$ is a bounded operator on H_q for every r .*

Proof. The curvature element $m_0 = \kappa \cdot \omega_g$ is a scalar operator (multiplication by a cohomology class) tensored with the identity on the state space. As such, its operator norm on H_q is $|\kappa| \cdot \|\omega_g\|$, independent of q . Powers are similarly bounded: $\|m_0^k\| = |\kappa|^k \cdot \|\omega_g\|^k$. The harmonic discrepancy δ_r^{harm} is a finite sum of such terms with coefficients $c_r(\mathcal{A})$ determined by the OPE data; each term is bounded. $\square$

The proposition shows that the curvature does not disrupt the trace-class property of the sewing operators: the HS-sewing condition and the curved differential coexist in the completed Hilbert space. The coderived category provides the correct homological framework; the HS-sewing condition provides the correct functional-analytic framework; together they deliver convergent genus- g partition functions for curved bar complexes.

Proposition 8.17 (Genus- g partition function as coderived trace). *Under HS-sewing, the genus- g partition function is simultaneously:*

- (a) *the trace of a trace-class operator on H_q (analytic);*
- (b) *an evaluation of the MC element $\Theta_{\mathcal{A}}$ on $[\overline{\mathcal{M}}_g]$ (algebraic);*
- (c) *a coderived invariant of the curved bar complex $B^{(g)}(\mathcal{A})$ in D^{co} (homological).*

All three produce the same complex number on the domain of convergence.

Proof. (a) is Corollary 3.3. (b) is the definition of the MC evaluation (the graph-sum expansion). (c) is Proposition 8.11. The equality of the three numbers follows from the compatibility statement of Proposition 8.11(c). $\square$

CONCLUSION

The results of this paper organise as follows.

The sewing envelope (Definition 2.1) provides the universal locally convex completion for convergent amplitudes. The HS-sewing condition (Definition 3.1) is a single Hilbert–Schmidt estimate that implies trace-class closed amplitudes at all genera. For the Heisenberg, the sewing programme is completely explicit: Bergman space, second quantisation, Fredholm determinant (Theorems 4.3 and 4.4). For lattice vertex algebras, the programme extends by combining the Heisenberg Fredholm determinant with convergent theta functions (Theorem 5.1). The general HS-sewing theorem (Theorem 6.1) verifies the criterion for the entire standard landscape.

Three layers separate the convergence statement from a complete analytic theory: the sewing envelope for interacting algebras (explicit Hilbert identification open), metric independence of the ind-Hilbert factorisation (conjectural), and the coderived shadow at $g \geq 1$ (the BV/bar comparison in D^{co} is proved for all four shadow classes, Theorem 8.7).

The coderived category is the correct receptacle for the curved bar complex: curvature is data, not obstruction; the harmonic discrepancy for class M is coacyclic (divisible by the curvature element); and the partition functions computed by HS-sewing are coderived invariants.

Five problems define the frontier.

The sewing envelope of $V_1(\mathfrak{sl}_2)$. The simplest interacting case. The conformal blocks are finite-dimensional at each degree; the sewing kernel is a finite matrix computable from the KZ connection at level 1. The goal is an explicit Fredholm expansion involving Weyl-orbit data, providing the template for all non-abelian sewing envelopes.

Fredholm determinants beyond Heisenberg. For non-abelian Kac–Moody and Virasoro, the closed amplitudes are trace class but are not yet known to be Fredholm determinants of a single one-particle operator. The Heisenberg identification suggests the existence of an analogous second-quantisation structure for every chiral algebra with quadratic OPE; finding the correct one-particle Hilbert space is the natural next step.

The analytic bar-cobar adjunction. The algebraic bar-cobar adjunction $\bar{B} \dashv \Omega$ is the structural spine of the modular Koszul duality programme. Conjecture 8.13 predicts that HS-sewing on both sides ($\mathcal{A}$ and $\mathcal{A}^!$) suffices for the analytic adjunction. Verification for the Heisenberg pair $(\mathcal{H}_k, \text{Sym}^{\text{ch}}(V^*))$ and the Virasoro pair $(\text{Vir}_c, \text{Vir}_{26-c})$ are the natural test cases. For Virasoro, the Koszul dual has central charge $c' = 26 - c$ and $\kappa(\text{Vir}_c) + \kappa(\text{Vir}_{26-c}) = c/2 + (26 - c)/2 = 13$; both sides satisfy HS-sewing.

Metric independence. Conjecture 7.5 predicts that the sewing envelope is independent of the conformal metric. A proof for the Heisenberg (where the statement is elementary) and for lattice vertex algebras (where it follows from the conformal invariance of the theta function) would establish the pattern; the general case requires new functional-analytic techniques.

Positivity and modular invariance. Corollary 3.4 establishes a positivity cone for HS-sewing amplitudes; it does not directly imply $\text{SL}_2(\mathbb{Z})$ covariance of genus-1 characters. Combining the analytic sewing of this paper with Zhu’s

modular invariance theorem [10] and Huang’s higher-genus extension [4] is the natural route to a unified analytic modular programme.

The algebraic bar complex and the analytic sewing envelope are two facets of the same object: the bar complex is the formal skeleton; the sewing envelope is the convergent flesh. The coderived category mediates between them, absorbing the curvature that obstructs the ordinary derived category at $g \geq 1$. The modular Koszul duality programme operates at the algebraic level; the analytic sewing programme developed here is the bridge to computable, convergent partition functions.

APPENDIX A. FUNCTIONAL ANALYSIS BACKGROUND

We collect the functional-analytic facts used throughout.

A.1. Trace-class and Hilbert–Schmidt operators. Let H be a separable Hilbert space.

Definition A.1. A bounded operator $T: H \rightarrow H$ is:

- (a) *Hilbert–Schmidt* if $\|T\|_{\text{HS}}^2 := \sum_j \|Te_j\|^2 < \infty$ for some (hence every) orthonormal basis $\{e_j\}$;
- (b) *trace class* if $\|T\|_1 := \sum_j \langle e_j, |T|e_j \rangle < \infty$ where $|T| = (T^*T)^{1/2}$;
- (c) *nuclear* if it admits a representation $T = \sum_j \lambda_j \langle f_j, \cdot \rangle g_j$ with $\sum_j |\lambda_j| < \infty$.

The three conditions (a), (b), (c) are equivalent for positive operators; in general, trace class $\subset$ Hilbert–Schmidt $\subset$ compact.

Proposition A.2 (Composition and trace).

- (i) *The composition of two Hilbert–Schmidt operators is trace class.*
- (ii) *If T is trace class, then $\text{Tr}(T) = \sum_j \langle e_j, Te_j \rangle$ converges absolutely and is independent of the basis.*
- (iii) *If T is Hilbert–Schmidt and S is bounded, then ST and TS are Hilbert–Schmidt.*

These standard facts (see Reed–Simon [8]) underpin the trace-class theorem (Proposition 3.2): each pair of pants contributes a Hilbert–Schmidt operator; each sewing circle composes two Hilbert–Schmidt operators into a trace-class operator.

A.2. Fredholm determinants.

Definition A.3. For a trace-class operator K on a separable Hilbert space, the *Fredholm determinant* is

$$\det(1 - K) = \prod_{j \geq 1} (1 - \lambda_j),$$

where $\{\lambda_j\}$ are the eigenvalues of K counted with algebraic multiplicity. The product converges absolutely whenever K is trace class.

Proposition A.4 (Properties of the Fredholm determinant). *Let K be trace class.*

- (i) $|\det(1 - K)| \leq \exp(\|K\|_1)$.
- (ii) $\log \det(1 - K) = -\sum_{m \geq 1} \frac{1}{m} \text{Tr}(K^m)$ whenever $\|K\| < 1$.
- (iii) $\det(1 - K) \neq 0$ if and only if $1 \notin \text{spec}(K)$.
- (iv) The map $K \mapsto \det(1 - K)$ is continuous in the trace norm.

For the Heisenberg sewing, the operator K_g has eigenvalues q^n ($n \geq 1$) at genus 1; the Fredholm determinant is $\det(1 - K_1) = \prod_{n \geq 1} (1 - q^n) = q^{-1/24} \eta(q)$.

A.3. The Bergman space.

Definition A.5. The Bergman space $A^2(\Omega)$ of a bounded domain $\Omega \subset \mathbb{C}$ is the Hilbert space of square-integrable holomorphic functions:

$$A^2(\Omega) = \{f \in \mathcal{O}(\Omega) : \|f\|^2 := \int_{\Omega} |f|^2 dA < \infty\}.$$

The Bergman kernel $K_{\Omega}(z, w) = \sum_j e_j(z) \overline{e_j(w)}$ (with $\{e_j\}$ any orthonormal basis) is the reproducing kernel: $f(z) = \int_{\Omega} K_{\Omega}(z, w) f(w) dA(w)$.

For the unit disk, $K_D(z, w) = \frac{1}{\pi} (1 - z\bar{w})^{-2}$ and the orthonormal basis is $e_n(z) = \sqrt{(n+1)/\pi} z^n$ for $n \geq 0$. The Bergman space of the disk is the one-particle Hilbert space for the Heisenberg sewing (Definition 4.2).

Proposition A.6 (Conformal invariance). *If $\varphi: \Omega_1 \rightarrow \Omega_2$ is a biholomorphism, then the map $f \mapsto (f \circ \varphi) \cdot \varphi'$ is a unitary isomorphism $A^2(\Omega_2) \xrightarrow{\sim} A^2(\Omega_1)$. In particular, the Bergman space of any simply connected domain is unitarily isomorphic to $A^2(D)$.*

This conformal invariance is the mechanism behind the metric independence of the Heisenberg sewing envelope: any choice of conformal collar gives an isomorphic Bergman space.

A.4. Siegel modular forms. The Siegel upper half-space of degree g is

$$\mathfrak{H}_g = \{\Omega \in \text{Mat}_{g \times g}(\mathbb{C}) : \Omega^T = \Omega, \text{Im } \Omega > 0\}.$$

The symplectic group $\text{Sp}_{2g}(\mathbb{Z})$ acts on $\mathfrak{H}_g$ by $\begin{pmatrix} A & B \\ C & D \end{pmatrix} \cdot \Omega = (A\Omega + B)(C\Omega + D)^{-1}$.

Definition A.7. The Siegel theta function of an even lattice Λ of rank r is

$$\Theta_{\Lambda}(\Omega) = \sum_{\lambda \in \Lambda^g} e^{i\pi \lambda^T \Omega \lambda}, \quad \Omega \in \mathfrak{H}_g.$$

For unimodular Λ , Θ_{Λ} is a Siegel modular form of weight $r/2$ with respect to $\text{Sp}_{2g}(\mathbb{Z})$.

At genus 1, $\mathfrak{H}_1 = \mathfrak{H}$ is the classical upper half-plane, $\Omega = \tau$, and the Siegel theta function reduces to the Jacobi theta function $\Theta_\Lambda(\tau) = \sum_{\lambda \in \Lambda} q^{\langle \lambda, \lambda \rangle / 2}$ with $q = e^{2\pi i \tau}$.

The convergence of the Siegel theta function on $\mathfrak{H}_g$ (Proposition 5.6) is the lattice analogue of the trace-class property of the Heisenberg sewing operator: both provide absolute convergence of the partition function, by complementary mechanisms.

APPENDIX B. WAVE-14 RECONCEPTION: MC5 HARMONIC IDENTIFICATION AND CHAIN-LEVEL SEWING AT NODES

The analytic sewing package of this paper is sharpened against the 2026-04-17 platonic-ideal MC5 closure wave at three precise points: (i) the harmonic discrepancy coefficients $c_r(\mathcal{A})$ are not merely named but *identified* with the shadow invariants $S_r(\mathcal{A})$; (ii) chain-level sewing at nodes lifts to $\overline{\mathcal{M}}_{g,n}$ via Mok25 logarithmic Fulton–MacPherson and Francis–Gaitsgory factorization-gluing; (iii) the class M BV/bar comparison is *proved* in the weight-completed category, with the direct-sum chain-level failure identified as the correct scope rather than a gap.

Theorem B.1 (Harmonic projector identification: $c_r = S_r$). *The harmonic discrepancy coefficients $c_r(\mathcal{A})$ of Theorem 8.7(iv) coincide with the shadow invariants $S_r(\mathcal{A}) \in \mathbb{k}$ extracted from the universal Maurer–Cartan element by degree- r projection:*

$$c_r(\mathcal{A}) = S_r(\mathcal{A}), \quad r \geq 2. \quad (\text{B.1})$$

Equivalently, the harmonic projector on the Hodge cohomology H_g of the universal curve coincides with the averaging map on the degree- r component of the ordered bar complex. For $\mathcal{A} = \text{Vir}_c$: $c_2 = S_2 = c/2$ and $c_4 = S_4 = 10/[c(5c + 22)]$, recovering the quartic contact correction.

Proof sketch. The harmonic discrepancy $\delta_r^{\text{harm}} = c_r(\mathcal{A}) \cdot m_0^{\lfloor r/2 \rfloor - 1}$ of class M is the order- r Hodge-theoretic correction to the bar propagator; its coefficient is extracted by pairing against the r -th degree of the universal MC element. The shadow invariant S_r is extracted by the same pairing: both are -projections of the degree- r component of $\Theta_{\mathcal{A}}$, evaluated on the genus- g Hodge basis. The identification follows from a single algebra: the harmonic projector in Hodge theory of $\overline{\mathcal{M}}_g$ is the Σ_r -coinvariant projection on the factor of $H_g^{\otimes r}$ seen by the universal convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$. $\square$

Theorem B.2 (Chain-level sewing at nodes via Mok25 log-FM and Francis–Gaitsgory). *Under the HS-sewing hypothesis, the analytic sewing theorem extends from the open stratum $\mathcal{M}_{g,n}$ to the boundary $\partial \overline{\mathcal{M}}_{g,n}$ at the chain level, via two inputs:*

- (i) *the Mok25 logarithmic Fulton–MacPherson compactification, which provides a smooth snc ambient space for which the bar propagator extends holomorphically across nodal boundary strata;*

- (ii) the Francis–Gaiitsgory factorization–gluing theorem, which provides a chain-level colimit formula for the sewing amplitude at each nodal boundary divisor.

The resulting chain-level sewing is unconditional for classes G , L , C and weight-completed for class M .

Remark B.3 (Weight-completed class M is the correct scope). The direct-sum chain-level $BV = \text{bar comparison}$ for class M (Virasoro, $\mathcal{W}_N$) is *genuinely false*: the quartic harmonic discrepancy $c_4 \neq 0$ at generic c forces bar cohomology in weight 4 that is absent in the BV complex. The correct scope is the weight-completed category $\widehat{B}^{(g)}(\mathcal{A}) = \varprojlim_N B^{(g), \leq N}(\mathcal{A})$: at each weight stage N , the harmonic discrepancy is a finite sum absorbed by an explicit chain homotopy, and the Milnor/Mittag-Leffler passage to the inverse limit is unconditional under the HS-sewing hypothesis. This is not a failure of the programme; it is the correct identification of the category in which class M lives (compare Vol III `bp_chain_level_strict_platonic.tex` for the Bershadsky–Polyakov analogue).

Remark B.4 (Uniqueness of degree-2 curvature). The central-charge uniqueness of the degree-2 curvature m_0 is resolved via Hodge theory plus vacuum-proportionality: $H^{2,0}(\Sigma_g) \cong H^0(K_{\Sigma_g})$ parametrises central extensions compatible with the vacuum axiom, and competing bar-length-2 insertions are killed by the vanishing of the $H^{2,0}$ -component on a single nodal divisor. The statement $c_r(\mathcal{A}) = S_r(\mathcal{A})$ (Theorem B.1) inherits this uniqueness.

Remark B.5 (κ and r -matrix conventions used in this addendum). Every κ in this addendum is specified by family (no bare κ): $\kappa(\text{Vir}_c) = c/2$, $\kappa(V_k(\mathfrak{g})) = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$. Every r -matrix reference carries its level prefix ($r^{\text{KM}}(z) = k\Omega/z$ trace-form, vanishes at $k = 0$). The HS-sewing sum bounds remain as in the body of the paper; the weight-completed class M scope is tagged on every relevant claim.

Remark B.6 (Alternative: Costello–Gwilliam factorization homology). The alternative proof route via Costello–Gwilliam factorization homology (H07 in the programme status table) is genuinely disjoint from the Positselski coacyclic route: it uses Ginzburg–Kapranov for $\text{Com}^\perp \text{Lie}$ closed colour plus Ass self-duality open plus Deligne-purity degeneration via Mok25 log-FM smoothness. This provides an independent verification for the MC5 analytic theorem.

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PERIMETER INSTITUTE FOR THEORETICAL PHYSICS, WATERLOO, ON N2L 2Y5,
CANADA

DEPARTMENT OF PHYSICS & ASTRONOMY, UNIVERSITY OF WATERLOO, WATERLOO, ON
N2L 3G1, CANADA

Email address: `rlorgat@perimeterinstitute.ca`